Validation and safety support for CareAI

April 10, 2026

Research Goal: Find academic papers that can strengthen the validation, safety, and deployment framing of CareAI / HAN++, an end-to-end clinical decision support system for multi-label binary disease prediction from EHR data with physician-in-the-loop confirmatory test recommendation. The search should focus on prior work relevant to unseen or external patient validation for diagnosis prediction, especially cases with small external cohorts; appropriate evaluation and reporting of discrimination and threshold-dependent metrics in such settings; uncertainty quantification for clinical prediction using MC dropout or related Bayesian approximations; and human-in-the-loop deferral or physician review workflows for uncertain predictions. The search should also find academic prior work relevant to the second-stage confirmatory test recommendation and iterative refinement protocol described for CareAI, where targeted additional tests are recommended for uncertain cases and predictions are updated after new results are available. The goal is to identify papers we should cite to support these claims and design choices, as well as comparison papers or baselines that would strengthen the manuscript's discussion of generalisation, safety, and clinical deployment.

Found 165 papers · April 10, 2026

Paper Catalog (165 papers)

	Year	Cit/yr	Title	Authors	Journal
1	2017	148	Discrimination and Calibration of Clinical Prediction Models: Users' Guides to the Medical Literature (link)	A. Alba et al.	JAMA
2	2020	327	An overview of clinical decision support systems: benefits, risks, and strategies for success (link)	Reed Taylor Sutton et al.	NPJ Digital Medicine
3	2019	18	Analyzing the role of model uncertainty for electronic health records (link)	Michael W. Dusenberry et al.	Proceedings of the ACM Conference on Health, Inference, and Learning
4	2016	89	Net benefit approaches to the evaluation of prediction models, molecular markers, and diagnostic tests (link)	A. Vickers, B. Van calster, and E. Steyerberg	The BMJ
5	2015	21	Transparent reporting of a multivariable prediction model for individual prognosis or diagnosis (TRIPOD): the TRIPOD Statement (link)	G. Collins, Johannes B Reitsma, Douglas G. Altman, and K. Moons	BMC Medicine

	Year	Cit/yr	Title	Authors	Journal
6	2021	51	Minimum sample size for external validation of a clinical prediction model with a binary outcome (link)	R. Riley et al.	Statistics in Medicine
7	2021	74	Second opinion needed: communicating uncertainty in medical machine learning (link)	Benjamin Kompa, Jasper Snoek, and A. Beam	NPJ Digital Medicine
8	2021	4.7	Incorporating uncertainty in learning to defer algorithms for safe computer-aided diagnosis (link)	Jessie Liu, B. Gallego, and S. Barbieri	Scientific Reports
9	2019	121	A simple, step-by-step guide to interpreting decision curve analysis (link)	A. Vickers, B. Van calster, and E. Steyerberg	Diagnostic and Prognostic Research
10	2016	52	Assessing the Clinical Impact of Risk Prediction Models With Decision Curves: Guidance for Correct Interpretation and Appropriate Use. (link)	Kathleen F. Kerr, Marshall D. Brown, Kehao Zhu, and H. Janes	Journal of clinical oncology : official journal of the American Society of Clinical Oncology
11	2019	233	PROBAST: A Tool to Assess the Risk of Bias and Applicability of Prediction Model Studies (link)	R. Wolff et al.	Annals of Internal Medicine
12	2020	118	External validation of prognostic models: what, why, how, when and where? (link)	Chava L. Ramspek, K. Jager, F. Dekker, C. Zoccali, and M. van Diepen	Clinical Kidney Journal
13	2023	56	There is no such thing as a validated prediction model (link)	B. Van calster, E. Steyerberg, L. Wynants, and M. van Smeden	BMC Medicine
14	2006	211	Decision Curve Analysis: A Novel Method for Evaluating Prediction Models (link)	A. Vickers and E. Elkin	Medical Decision Making
15	2018	126	Reporting and Interpreting Decision Curve Analysis: A Guide for Investigators. (link)	B. Van calster et al.	European urology
16	2022	30	Tackling prediction uncertainty in machine learning for healthcare (link)	Michelle Chua et al.	Nature Biomedical Engineering

	Year	Cit/yr	Title	Authors	Journal
17	2024	50	Evaluation of clinical prediction models (part 3): calculating the sample size required for an external validation study (link)	R. Riley et al.	The BMJ
18	2021	17	External validation of clinical prediction models: simulation-based sample size calculations were more reliable than rules-of-thumb (link)	Kym I. E. Snell et al.	Journal of Clinical Epidemiology
19	2021	65	Risk of bias in studies on prediction models developed using supervised machine learning techniques: systematic review (link)	C. A. Andaur Navarro et al.	The BMJ
20	2019	3.3	Statistical inference for net benefit measures in biomarker validation studies (link)	T. Marsh, H. Janes, and M. Pepe	Biometrics
21	1992	4.6	What price perfection? Calibration and discrimination of clinical prediction models. (link	G. Diamond	Journal of clinical epidemiology
22	2020	7.4	Deep Bayesian Gaussian processes for uncertainty estimation in electronic health records (link)	Yikuan Li et al.	Scientific Reports
23	2021	10	Estimation of required sample size for external validation of risk models for binary outcomes (link)	M. Pavlou et al.	Statistical Methods in Medical Research
24	2019	147	PROBAST: A Tool to Assess Risk of Bias and Applicability of Prediction Model Studies: Explanation and Elaboration (link)	K. Moons et al.	Annals of Internal Medicine
25	2022	9.1	Calibrated Selective Classification (link)	Adam Fisch, T. Jaakkola, and R. Barzilay	Trans. Mach. Learn. Res.
26	2018	19	Direct Uncertainty Prediction for Medical Second Opinions (link)	M. Raghu et al.	International Conference on Machine Learning
27	2020	16	Trust Issues: Uncertainty Estimation Does Not Enable Reliable OOD Detection On Medical Tabular Data (link)	Dennis Ulmer, L. Meijerink, and G. Ciná	ML4H@NeurIPS

	Year	Cit/yr	Title	Authors	Journal
28	2023	0.9	Safe and reliable transport of prediction models to new healthcare settings without the need to collect new labeled data (link)	Rudraksh Tuwani and Andrew L. Beam	medRxiv
29	2018	12	Joint Active Feature Acquisition and Classification with Variable-Size Set Encoding (link)	Hajin Shim, Sung Ju Hwang, and Eunho Yang	Neural Information Processing Systems
30	2018	4.2	An Optimal Policy for Patient Laboratory Tests in Intensive Care Units (link)	Li-Fang Cheng, Niranjani Prasad, and B. Engelhardt	Pacific Symposium on Biocomputing. Pacific Symposium on Biocomputing
31	2022	0.5	Confidence-based laboratory test reduction recommendation algorithm (link)	Ting Huang, L. Li, E. Bernstam, and X. Jiang	BMC Medical Informatics and Decision Making
32	2024	10	AI-Driven Diagnostic Assistance in Medical Inquiry: Reinforcement Learning Algorithm Development and Validation (link)	Xuan Zou et al.	Journal of Medical Internet Research
33	2024	1.7	Extended sample size calculations for evaluation of prediction models using a threshold for classification (link)	Rebecca Whittle et al.	BMC Medical Research Methodology
34	2021	41	The importance of being external. methodological insights for the external validation of machine learning models in medicine (link)	F. Cabitza et al.	Computer methods and programs in biomedicine
35	2020	17	Minimum sample size for external validation of a clinical prediction model with a continuous outcome (link)	Lucinda Archer et al.	Statistics in Medicine
36	2022	22	Targeted validation: validating clinical prediction models in their intended population and setting (link)	M. Sperrin, R. Riley, G. Collins, and G. Martin	Diagnostic and Prognostic Research
37	2022	32	Stability of clinical prediction models developed using statistical or machine learning methods (link)	R. Riley and G. Collins	Biometrical Journal. Biometrische Zeitschrift
38	2021	2.3	BEDS-Bench: Behavior of EHR-models under Distributional Shift-A Benchmark (link)	A. Avati et al.	ArXiv

	Year	Cit/yr	Title	Authors	Journal
39	2021	47	Decision curve analysis to evaluate the clinical benefit of prediction models. (link)	A. Vickers and F. Holland	The spine journal : official journal of the North American Spine Society
40	2011	9.6	Decision curve analysis revisited: overall net benefit, relationships to ROC curve analysis, and application to case-control studies (link)	V. Rousson and T. Zumbrunn	BMC Medical Informatics and Decision Making
41	2022	36	Methodological guidance for the evaluation and updating of clinical prediction models: a systematic review (link)	M. A. Binuya, E. Engelhardt, W. Schats, M. Schmidt, and E. Steyerberg	BMC Medical Research Methodology
42	2025	8.6	Instability of the AUROC of Clinical Prediction Models (link)	Florian D van Leeuwen et al.	Statistics in Medicine
43	2016	43	Risk Prediction with Electronic Health Records: A Deep Learning Approach (link)	Yu Cheng, Fei Wang, Ping Zhang, and Jianying Hu	SDM
44	2019	1.0	Effective Medical Test Suggestions Using Deep Reinforcement Learning (link)	Yang Chen, Kai-Fu Tang, Yu-Shao Peng, and Edward Y. Chang	ArXiv
45	2023	7.3	Deep Reinforcement Learning for Cost-Effective Medical Diagnosis (link)	Zheng Yu et al.	ArXiv
46	2020	17	Clinical Implementation of Predictive Models Embedded within Electronic Health Record Systems: A Systematic Review (link)	Terrence C. Lee, N. Shah, Alyssa Haack, and Sally L. Baxter	Informatics (MDPI)
47	2019	2.8	Selective prediction-set models with coverage guarantees (link)	Jean Feng, A. Sondhi, Jessica Perry, and N. Simon	ArXiv
48	2019	86	Conformal Prediction Under Covariate Shift (link)	R. Tibshirani, R. Barber, E. Candès, and Aaditya Ramdas	Neural Information Processing Systems
49	2020	2.5	CaliForest: calibrated random forest for health data (link)	Yubin Park and Joyce Ho	Proceedings of the ACM Conference on Health, Inference, and Learning

	Year	Cit/yr	Title	Authors	Journal
50	2021	2.3	Uncertainty-Aware Time-to-Event Prediction using Deep Kernel Accelerated Failure Time Models (link)	Zhiliang Wu, Yinchong Yang, P. Fasching, and Volker Tresp	Machine Learning in Health Care
51	2022	0.6	Deep Kernel Learning for Mortality Prediction in the Face of Temporal Shift (link)	Miguel Rios and A. Abu-Hanna	ArXiv
52	2007	18	Some Unintended Consequences of Clinical Decision Support Systems (link)	J. Ash, Dean F. Sittig, Emily M. Campbell, Kenneth P. Guappone, and R. H. Dykstra	AMIA ... Annual Symposium proceedings. AMIA Symposium
53	2021	4.8	A survey of extant organizational and computational setups for deploying predictive models in health systems (link)	Sehj Kashyap, K. Morse, Birju S. Patel, and N. Shah	Journal of the American Medical Informatics Association : JAMIA
54	2021	15	Adoption of clinical risk prediction tools is limited by a lack of integration with electronic health records (link)	Videha Sharma et al.	BMJ Health & Care Informatics
55	2016	22	Clinical Decision Support: a 25 Year Retrospective and a 25 Year Vision (link)	Blackford Middleton, D. Sittig, and A. Wright	Yearbook of Medical Informatics
56	2023	35	Transparent reporting of multivariable prediction models for individual prognosis or diagnosis: checklist for systematic reviews and meta-analyses (TRIPOD-SRMA) (link)	Kym I. E. Snell et al.	The BMJ
57	2022	23	Methodological conduct of prognostic prediction models developed using machine learning in oncology: a systematic review (link)	P. Dhiman et al.	BMC Medical Research Methodology
58	2021	7.5	Large-scale validation of the Prediction model Risk Of Bias ASsessment Tool (PROBAST) using a short form: high risk of bias models show poorer discrimination (link)	E. Venema et al.	Journal of clinical epidemiology

	Year	Cit/yr	Title	Authors	Journal
59	2023	18	Sample size requirements are not being considered in studies developing prediction models for binary outcomes: a systematic review (link)	P. Dhiman et al.	BMC Medical Research Methodology
60	2025	1.8	A tailored fit that doesn’t fit all: the problem of threshold overfitting in diagnostic studies (link)	J. Arredondo Montero	Diagnosis
61	2016	6.8	Using the weighted area under the net benefit curve for decision curve analysis (link)	R. Talluri and S. Shete	BMC Medical Informatics and Decision Making
62	2023	1.9	Bayesian Decision Curve Analysis With Bayesdca (link)	Giuliano Netto Flores Cruz and Keegan D. Korthauer	Statistics in Medicine
63	2015	0.5	Assessing calibration in an external validation study. (link)	G. Collins, E. Ogundimu, and Y. Le Manach	The spine journal : official journal of the North American Spine Society
64	2022	12	Post-hoc estimators for learning to defer to an expert (link)	Harikrishna Narasimhan, Wittawat Jitkrittum, A. Menon, A. Rawat, and Surinder Kumar	Advances in Neural Information Processing Systems 35
65	2021	35	Machine learning with a reject option: a survey (link)	Kilian Hendrickx, Lorenzo Perini, Dries Van der Plas, Wannes Meert, and Jesse Davis	Machine Learning
66	2021	2.6	Learning-to-defer for sequential medical decision-making under uncertainty (link)	Shalmali Joshi, S. Parbhoo, and F. Doshi-Velez	Trans. Mach. Learn. Res.
67	2019	53	SelectiveNet: A Deep Neural Network with an Integrated Reject Option (link)	Yonatan Geifman and Ran El-Yaniv	International Conference on Machine Learning
68	2016	2.4	EVALUATING RISK-PREDICTION MODELS USING DATA FROM ELECTRONIC HEALTH RECORDS. (link)	Le Wang, P. Shaw, H. Mathelier, S. Kimmel, and B. French	The annals of applied statistics

	Year	Cit/yr	Title	Authors	Journal
69	2020	8.5	Active Feature Acquisition with Generative Surrogate Models (link)	Yang Li and Junier B. Oliva	International Conference on Machine Learning
70	2023	1.4	Graph-based clinical recommender: Predicting specialists procedure orders using graph representation learning (link)	S. Fouladvand et al.	Journal of biomedical informatics
71	2021	4.9	Deep into Laboratory: An Artificial Intelligence Approach to Recommend Laboratory Tests (link)	M. Islam, T. N. Poly, Hsuan-Chia Yang, and Y. Li	Diagnostics
72	2025	5.9	Towards Trustworthy AI in Healthcare: Epistemic Uncertainty Estimation for Clinical Decision Support (link)	Adrian Lindenmeyer, Malte Blattmann, S. Franke, T. Neumuth, and Daniel Schneider	Journal of Personalized Medicine
73	2023	156	Conformal Prediction: A Gentle Introduction (link)	Anastasios Nikolas Angelopoulos and Stephen Bates	Found. Trends Mach. Learn.
74	2020	12	Probabilistic Machine Learning for Healthcare (link)	I. Chen, Shalmali Joshi, M. Ghassemi, and R. Ranganath	Annual review of biomedical data science
75	2020	40	Efficient and Scalable Bayesian Neural Nets with Rank-1 Factors (link)	Michael W. Dusenberry et al.	ArXiv
76	2024	1.4	Inadequacy of common stochastic neural networks for reliable clinical decision support (link)	Adrian Lindenmeyer, Malte Blattmann, S. Franke, T. Neumuth, and Daniel Schneider	ArXiv
77	2022	7.5	Conformal Prediction Sets with Limited False Positives (link)	Adam Fisch, Tal Schuster, T. Jaakkola, and R. Barzilay	ArXiv
78	2019	9.1	Federated Uncertainty-Aware Learning for Distributed Hospital EHR Data (link)	Sabri Boughorbel et al.	ArXiv
79	2020	77	Classification with Valid and Adaptive Coverage (link)	Yaniv Romano, Matteo Sesia, and E. Candès	arXiv: Methodology

	Year	Cit/yr	Title	Authors	Journal
80	2020	79	Uncertainty Sets for Image Classifiers using Conformal Prediction (link)	Anastasios Nikolas Angelopoulos, Stephen Bates, J. Malik, and Michael I. Jordan	ArXiv
81	2025		Presenting Artificial Intelligence Predictions Based on Electronic Medical Records to Clinicians in Hospitals: A Systematic Review (link)	Monica Noselli et al.	IEEE Journal of Biomedical and Health Informatics
82	2012	41	Health IT and Patient Safety: Building Safer Systems for Better Care (link)	D. Chou	JAMA
83	2012	7.6	The dangerous decade (link)	E. Coiera, J. Aarts, and C. Kulikowski	Journal of the American Medical Informatics Association : JAMIA
84	2020	2.7	Development, Implementation, and Evaluation of a Personalized Machine Learning Algorithm for Clinical Decision Support: Case Study With Shingles Vaccination (link)		Journal of Medical Internet Research
85	2024	13	Factors influencing clinician and patient interaction with machine learning-based risk prediction models: a systematic review. (link	R. Giddings et al.	The Lancet. Digital health
86	2001	29	Clinical decision support systems for the practice of evidence-based medicine. (link)	I. Sim et al.	Journal of the American Medical Informatics Association : JAMIA
87	2015	78	Transparent reporting of a multivariable prediction model for individual prognosis or diagnosis (TRIPOD): The TRIPOD statement (link)	G. Collins, J. Reitsma, DG Altman, and K. Moons	British Journal of Cancer
88	2020	8.5	Transparent Reporting of Multivariable Prediction Models in Journal and Conference Abstracts: TRIPOD for Abstracts (link)	P. Heus et al.	Annals of Internal Medicine
89	2015	53	Transparent reporting of a multivariable prediction model for Individual Prognosis or Diagnosis (TRIPOD): the TRIPOD statement. (link)	Gary S. Collins, J. Reitsma, D. Altman, and K. Moons	Journal of clinical epidemiology

	Year	Cit/yr	Title	Authors	Journal
90	2019	12	Uniformity in measuring adherence to reporting guidelines: the example of TRIPOD for assessing completeness of reporting of prediction model studies (link)	P. Heus et al.	BMJ Open
91	2020	9.7	Ambiguity-aware AI Assistants for Medical Data Analysis (link)	M. Schaekermann et al.	Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems
92	2022	4.1	Selective prediction for extracting unstructured clinical data (link)	Akshay Swaminathan et al.	Journal of the American Medical Informatics Association : JAMIA
93	2018	1.2	Direct Uncertainty Prediction with Applications to Healthcare (link)	M. Raghu et al.	ArXiv
94	2020	11	A framework for making predictive models useful in practice (link)	Kenneth Jung et al.	Journal of the American Medical Informatics Association : JAMIA
95	2010	26	On the Foundations of Noise-free Selective Classification (link)	Ran El-Yaniv and Yair Wiener	J. Mach. Learn. Res.
96	2025		Is Uncertainty Quantification a Viable Alternative to Learned Deferral? (link)	Anna M. Wundram and Christian F. Baumgartner	ArXiv
97	2025		On the Limits of Selective AI Prediction: A Case Study in Clinical Decision Making (link)	Sarah Jabbour et al.	ArXiv
98	2023	4.2	Embracing the uncertainty in human–machine collaboration to support clinical decision-making for mental health conditions (link)	Ram Popat and Julia Ive	Frontiers in Digital Health
99	2020	1.7	Diagnostic Uncertainty Calibration: Towards Reliable Machine Predictions in Medical Domain (link)	Takahiro Mimori, Keiko Sasada, H. Matsui, and Issei Sato	International Conference on Artificial Intelligence and Statistics
100	2024	1.8	Expected Value of Sample Information Calculations for Risk Prediction Model Validation (link)	M. Sadatsafavi, Andrew J. Vickers, Tae Yoon Lee, Paul Gustafson, and Laure Wynants	Medical Decision Making

	Year	Cit/yr	Title	Authors	Journal
101	2021	16	Radiomics machine learning study with a small sample size: Single random training-test set split may lead to unreliable results (link)	Chansik An et al.	PLoS ONE
102	2016	3.4	Risk Prediction With Electronic Health Records: The Importance of Model Validation and Clinical Context. (link)	B. Goldstein, A. Navar, and M. Pencina	JAMA cardiology
103	2022	3.3	Value-of-Information Analysis for External Validation of Risk Prediction Models (link)	M. Sadatsafavi, Tae Yoon Lee, Laure Wynants, A. Vickers, and P. Gustafson	Medical Decision Making
104	2021	17	Minimum sample size calculations for external validation of a clinical prediction model with a time-to-event outcome (link)	R. Riley et al.	Statistics in Medicine
105	2025		Sample size determination for external validation of risk models with binary outcomes using the area under the ROC curve (link)	Yiwang Zhou, Ying Zheng, Cai Li, Akshay Sharma, and Li Tang	Epidemiologic Methods
106	2006	0.2	Active Feature Acquisition with POMDP Models (link)	Qi An, Hui Li, X. Liao, and L. Carin	
107	2021	1.5	Learning a Diagnostic Strategy on Medical Data With Deep Reinforcement Learning (link)	Mengxia Zhu and Haogang Zhu	IEEE Access
108	2019	5.5	Classification with costly features as a sequential decision-making problem (link)	Jaromír Janisch, T. Pevný, and V. Lisý	Machine Learning
109	2016	6.3	Predicting inpatient clinical order patterns with probabilistic topic models vs conventional order sets (link)	Jonathan H. Chen, M. Goldstein, S. Asch, Lester W. Mackey, and R. Altman	Journal of the American Medical Informatics Association : JAMIA
110	2009	1.6	Breaking boundaries: active information acquisition across learning and diagnosis (link)	Ashish Kapoor and E. Horvitz	Neural Information Processing Systems

	Year	Cit/yr	Title	Authors	Journal
111	2024	5.0	Deep Reinforcement Learning for Personalized Diagnostic Decision Pathways Using Electronic Health Records: A Comparative Study on Anemia and Systemic Lupus Erythematosus (link)	Lillian Muyama, A. Neuraz, and Adrien Coulet	Artificial intelligence in medicine
112	2023		Extracting Diagnosis Pathways from Electronic Health Records Using Deep Reinforcement Learning (link)	Lillian Muyama, A. Neuraz, and Adrien Coulet	ArXiv
113	2018	24	Decision curve analysis: a technical note. (link)	Wentao Bao et al.	Annals of translational medicine
114	2013	1.8	Flexible recalibration of binary clinical prediction models (link)	J. Dalton	Statistics in Medicine
115	2025		Decision curve analysis explained (link)	Javier Arredondo Montero	Diagnosis
116	2022		A Comparison of Uncertainty Estimation Methods for Deep Learning-based Clinical Risk Scores (link)	A. Srivastava, S. Shashikumar, and S. Nemati	American Medical Informatics Association Annual Symposium
117	2022	26	A Review on Bayesian Deep Learning in Healthcare: Applications and Challenges (link)	Abdullah A. Abdullah, M. M. Hassan, and Yaseen T. Mustafa	IEEE Access
118	2022	5.0	Modular Conformal Calibration (link)	Charles Marx, Shengjia Zhao, Willie Neiswanger, and Stefano Ermon	ArXiv
119	2007	78	A tutorial on conformal prediction (link)	G. Shafer and Vladimir Vovk	ArXiv
120	2020	0.2	Adaptive Prediction Timing for Electronic Health Records (link)	J. Deasy, A. Ercole, and Pietro Liò	ArXiv
121	2021	2.1	On Cost-Sensitive Calibrated Uncertainty in Deep Learning: An application on COVID-19 detection (link)	Biraja Ghoshal and A. Tucker	2021 IEEE 34th International Symposium on Computer-Based Medical Systems (CBMS)
122	2019	1.0	Modeling the Uncertainty in Electronic Health Records: a Bayesian Deep Learning Approach (link)	Riyi Qiu et al.	ArXiv

	Year	Cit/yr	Title	Authors	Journal
123	2020	4.5	Frequentist Uncertainty in Recurrent Neural Networks via Blockwise Influence Functions (link)	Ahmed M. Alaa and M. Schaar	International Conference on Machine Learning
124	2023	6.8	Robust Uncertainty Quantification using Conformalised Monte Carlo Prediction (link)	Daniel Bethell, Simos Gerasimou, and R. Calinescu	ArXiv
125	2019	2.1	Bayesian Modelling in Practice: Using Uncertainty to Improve Trustworthiness in Medical Applications (link)	David Ruhe, G. Ciná, Michele Tonutti, D. D. Bruin, and P. Elbers	ArXiv
126	2025	12	Uncertainty Quantification for Machine Learning in Healthcare: A Survey (link)	L. J. L. L'opez et al.	ACM Conference on Health, Inference, and Learning
127	2016	754	Simple and Scalable Predictive Uncertainty Estimation using Deep Ensembles (link)	Balaji Lakshminarayanan, A. Pritzel, and C. Blundell	Neural Information Processing Systems
128	2008	8.0	Clinical Decision Support Systems: A Review on Knowledge Representation and Inference Under Uncertainties (link)	G. Kong, Dongling Xu, and Jianbo Yang	Int. J. Comput. Intell. Syst.
129	2001	16	Evaluating informatics applications - clinical decision support systems literature review (link	B. Kaplan	International journal of medical informatics
130	2016	13	The Effects of Clinical Decision Support Systems on Medication Safety: An Overview (link)	P. Jia, Longhao Zhang, Jingjing Chen, Pujing Zhao, and Mingming Zhang	PLoS ONE
131	2012	33	Improving Outcomes with Clinical Decision Support: An Implementer's Guide (link)	M. J. A. Jerome A Osheroff et al.	
132	2021	7.3	Appraising prediction research: risk of bias and applicability assessment of prognostic models using the PROBAST. (link)	Y. de Jong et al.	Nephrology
133	2021	5.5	Appraising prediction research: a guide and meta-review on bias and applicability assessment using the Prediction model Risk Of Bias ASsessment Tool (PROBAST) (link)	Y. Jong et al.	Nephrology (Carlton, Vic.)

	Year	Cit/yr	Title	Authors	Journal
134	2025	8.3	Statistical primer: sample size considerations for developing and validating clinical prediction models (link)	Glen P. Martin, Richard D Riley, J. Ensor, and S. Grant	European Journal of Cardio-Thoracic Surgery
135	2024	7.6	Twelve practical recommendations for developing and applying clinical predictive models (link)	Guoshuang Feng et al.	The Innovation Medicine
136	2018	14	Poor reporting of multivariable prediction model studies: towards a targeted implementation strategy of the TRIPOD statement (link)	P. Heus et al.	BMC Medicine
137	2015	36	Transparent Reporting of a Multivariable Prediction Model for Individual Prognosis or Diagnosis (TRIPOD): the TRIPOD Statement (link)	G. Collins, Johannes B Reitsma, Douglas G. Altman, and K. Moons	Diabetic Medicine
138	2021	1.2	Learning decision thresholds for risk stratification models from aggregate clinician behavior (link)	Birju S. Patel, E. Steinberg, S. Pfohl, and N. Shah	Journal of the American Medical Informatics Association : JAMIA
139	2023	1.3	Human-Centered Deferred Inference: Measuring User Interactions and Setting Deferral Criteria for Human-AI Teams (link)	Stephan J. Lemmer, Anhong Guo, and Jason J. Corso	Proceedings of the 28th International Conference on Intelligent User Interfaces
140	2023	11	In Defense of Softmax Parametrization for Calibrated and Consistent Learning to Defer (link)	Yuzhou Cao, Hussein Mozannar, Lei Feng, Hongxin Wei, and Bo An	ArXiv
141	2023	33	Assertiveness-based Agent Communication for a Personalized Medicine on Medical Imaging Diagnosis (link)	F. M. Calisto et al.	Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems
142	2021	8.6	TRIPOD Reporting Guidelines for Diagnostic and Prognostic Studies. (link)	R. Patzer, A. Kaji, and Y. Fong	JAMA surgery
143	2021	2.9	Pre-emptive learning-to-defer for sequential medical decision-making under uncertainty (link)	Shalmali Joshi, S. Parbhoo, and F. Doshi-Velez	ArXiv
144	2023	2.2	Evaluating medical AI systems in dermatology under uncertain ground truth (link)	David Stutz et al.	Medical image analysis

	Year	Cit/yr	Title	Authors	Journal
145	2025	2.0	"I don't know": An uncertainty-aware machine learning model for predicting patient disposition at emergency department triage (link)	Abubakar Sadiq Bouda Abdulai, Jean Storm, and Michael Ehrlich	International journal of medical informatics
146		2	Learning to Defer with an Uncertain Rejector via Conformal Prediction (link)	Yizirui Fang and Eric Nalisnick	
147	2024	1.1	Selective classification with machine learning uncertainty estimates improves ACS prediction: A retrospective study in the prehospital setting (link)	J. J. Garcia, Rebecca Kitzmiller, Ashok K. Krishnamurthy, and J. Zègre-Hemsey	Research Square
148	2024	12	ADNEX risk prediction model for diagnosis of ovarian cancer: systematic review and meta-analysis of external validation studies (link)	L. Barreñada et al.	BMJ Medicine
149	2025	3.2	External validation of the COLOFIT colorectal cancer risk prediction model in the Oxford-FIT dataset: the importance of population characteristics and clinically relevant evaluation metrics (link)	A. Tamm et al.	BMC Medicine
150	2025	1.0	Bayesian Sample Size Calculations for External Validation Studies of Risk Prediction Models (link)	M. Sadatsafavi, Paul Gustafson, S. Setayeshgar, Laure Wynants, and Richard D Riley	Statistics in Medicine
151	2021	8.1	Moving beyond AUC: decision curve analysis for quantifying net benefit of risk prediction models (link)	M. Sadatsafavi et al.	European Respiratory Journal
152	2023	3.1	Improving Clinical Decision Making With a Two-Stage Recommender System (link)	Shaina Raza and Chen Ding	IEEE/ACM Transactions on Computational Biology and Bioinformatics
153	2025	4.9	Language Agents for Hypothesis-driven Clinical Decision Making with Reinforcement Learning (link)	David Bani-Harouni, Chantal Pellegrini, Ege Özsoy, Matthias Keicher, and Nassir Navab	ArXiv
154	1985	1.2	Diagnostic testing revisited: pathways through uncertainty. (link)	M. Schechter and S. Sheps	Canadian Medical Association journal

	Year	Cit/yr	Title	Authors	Journal
155	2024	2.8	ED-Copilot: Reduce Emergency Department Wait Time with Language Model Diagnostic Assistance (link)	Liwen Sun, Abhineet Agarwal, Aaron E. Kornblith, Bin Yu, and Chenyan Xiong	International Conference on Machine Learning
156	2019	1.3	Optimal testing policies for diagnosing patients with intermediary probability of disease (link)	E. Arruda, B. B. Pereira, C. A. Thiers, and B. Tura	Artificial intelligence in medicine
157	2020	3.7	Reinforcement Learning with Efficient Active Feature Acquisition (link)	Haiyan Yin, Yingzhen Li, Sinno Jialin Pan, Cheng Zhang, and Sebastian Tschiatschek	ArXiv
158	1980		On-Line Bayesian Decision Analysis in a Sequential Laboratory-Directed Diagnostic Protocol. (link)	E. Turpin and J. Beck	
159	2020	0.3	Learning to Ask Medical Questions using Reinforcement Learning (link)	Uri Shaham et al.	ArXiv
160	2018		Adaptive Sequential Laboratory Diagnostic Tests: Joint Bayesian Analysis for Optimality (link)	Zeyneb Guenfoud and P. Antal	
161	1999	0.1	Optimizing Diagnostic Test Sequences: The Probability Modifying Plot (link)	J. Severens, P. F. D. Vries Robbé, and A. Verbeek	Methods of Information in Medicine
162	2025		ExOSITO: Explainable Off-Policy Learning with Side Information for Intensive Care Unit Blood Test Orders (link)	Zongliang Ji, A. Amaral, Anna Goldenberg, and Rahul G. Krishnan	ArXiv
163	2018	0.1	A Model-Based Reinforcement Learning Approach for a Rare Disease Diagnostic Task (link)	R. Besson et al.	ArXiv
164	1997	1.5	Avoiding premature closure in sequential diagnosis (link	D. McSherry	Artificial intelligence in medicine
165	2020		Optimization of a Sequential Decision Making Problem for a Rare Disease Diagnostic Application (link)	R. Besson et al.	International Conference on Agents and Artificial Intelligence

Paper Details

1. · 2017 · 148 cit/yr

Discrimination and Calibration of Clinical Prediction Models: Users' Guides to the Medical Literature (link)

A. Alba et al.

JAMA · Oct 10, 2017 · 1258 citations

---

2. · 2020 · 327 cit/yr

An overview of clinical decision support systems: benefits, risks, and strategies for success (link)

Reed Taylor Sutton et al.

NPJ Digital Medicine · Feb 6, 2020 · 2018 citations

> Computerized clinical decision support systems, or CDSS, represent a paradigm shift in healthcare today. CDSS are used to augment clinicians in their complex decision-making processes. Since their first use in the 1980s, CDSS have seen a rapid evolution. They are now commonly administered through electronic medical records and other computerized clinical workflows, which has been facilitated by increasing global adoption of electronic medical records with advanced capabilities. Despite these advances, there remain unknowns regarding the effect CDSS have on the providers who use them, patient outcomes, and costs. There have been numerous published examples in the past decade(s) of CDSS success stories, but notable setbacks have also shown us that CDSS are not without risks. In this paper, we provide a state-of-the-art overview on the use of clinical decision support systems in medicine, including the different types, current use cases with proven efficacy, common pitfalls, and potential harms. We conclude with evidence-based recommendations for minimizing risk in CDSS design, implementation, evaluation, and maintenance.

---

3. · 2019 · 18 cit/yr

Analyzing the role of model uncertainty for electronic health records (link)

Michael W. Dusenberry et al.

Proceedings of the ACM Conference on Health, Inference, and Learning · Jun 10, 2019 · 121 citations

> In medicine, both ethical and monetary costs of incorrect predictions can be significant, and the complexity of the problems often necessitates increasingly complex models. Recent work has shown that changing just the random seed is enough for otherwise well-tuned deep neural networks to vary in their individual predicted probabilities. In light of this, we investigate the role of model uncertainty methods in the medical domain. Using RNN ensembles and various Bayesian RNNs, we show that population-level metrics, such as AUC-PR, AUC-ROC, log-likelihood, and calibration error, do not capture model uncertainty. Meanwhile, the presence of significant variability in patient-specific predictions and optimal decisions motivates the need for capturing model uncertainty. Understanding the uncertainty for individual patients is an area with clear clinical impact, such as determining when a model decision is likely to be brittle. We further show that RNNs with only Bayesian embeddings can be a more efficient way to capture model uncertainty compared to ensembles, and we analyze how model uncertainty is impacted across individual input features and patient subgroups.

---

4. · 2016 · 89 cit/yr

Net benefit approaches to the evaluation of prediction models, molecular markers, and diagnostic tests (link)

A. Vickers, B. Van calster, and E. Steyerberg

The BMJ · Jan 25, 2016 · 907 citations

> Many decisions in medicine involve trade-offs, such as between diagnosing patients with disease versus unnecessary additional testing for those who are healthy. Net benefit is an increasingly reported decision analytic measure that puts benefits and harms on the same scale. This is achieved by specifying an exchange rate, a clinical judgment of the relative value of benefits (such as detecting a cancer) and harms (such as unnecessary biopsy) associated with models, markers, and tests. The exchange rate can be derived by asking simple questions, such as the maximum number of patients a doctor would recommend for biopsy to find one cancer. As the answers to these sorts of questions are subjective, it is possible to plot net benefit for a range of reasonable exchange rates in a "decision curve." For clinical prediction models, the exchange rate is related to the probability threshold to determine whether a patient is classified as being positive or negative for a disease. Net benefit is useful for determining whether basing clinical decisions on a model, marker, or test would do more good than harm. This is in contrast to traditional measures such as sensitivity, specificity, or area under the curve, which are statistical abstractions not directly informative about clinical value. Recent years have seen an increase in practical applications of net benefit analysis to research data. This is a welcome development, since decision analytic techniques are of particular value when the purpose of a model, marker, or test is to help doctors make better clinical decisions.

---

5. · 2015 · 21 cit/yr

Transparent reporting of a multivariable prediction model for individual prognosis or diagnosis (TRIPOD): the TRIPOD Statement (link)

G. Collins, Johannes B Reitsma, Douglas G. Altman, and K. Moons

BMC Medicine · Jan 6, 2015 · 237 citations

> AbstractPrediction models are developed to aid health care providers in estimating the probability or risk that a specific disease or condition is present (diagnostic models) or that a specific event will occur in the future (prognostic models), to inform their decision making. However, the overwhelming evidence shows that the quality of reporting of prediction model studies is poor. Only with full and clear reporting of information on all aspects of a prediction model can risk of bias and potential usefulness of prediction models be adequately assessed. The Transparent Reporting of a multivariable prediction model for Individual Prognosis Or Diagnosis (TRIPOD) Initiative developed a set of recommendations for the reporting of studies developing, validating, or updating a prediction model, whether for diagnostic or prognostic purposes. This article describes how the TRIPOD Statement was developed. An extensive list of items based on a review of the literature was created, which was reduced after a Web-based survey and revised during a 3-day meeting in June 2011 with methodologists, health care professionals, and journal editors. The list was refined during several meetings of the steering group and in e-mail discussions with the wider group of TRIPOD contributors. The resulting TRIPOD Statement is a checklist of 22 items, deemed essential for transparent reporting of a prediction model

study. The TRIPOD Statement aims to improve the transparency of the reporting of a prediction model study regardless of the study methods used. The TRIPOD Statement is best used in conjunction with the TRIPOD explanation and elaboration document. To aid the editorial process and readers of prediction model studies, it is recommended that authors include a completed checklist in their submission (also available at www.tripod-statement.org). Editors' note: In order to encourage dissemination of the TRIPOD Statement, this article is freely accessible on the Annals of Internal Medicine Web site (www.annals.org) and will be also published in BJOG, British Journal of Cancer, British Journal of Surgery, BMC Medicine, British Medical Journal, Circulation, Diabetic Medicine, European Journal of Clinical Investigation, European Urology, and Journal of Clinical Epidemiology. The authors jointly hold the copyright of this article. An accompanying Explanation and Elaboration article is freely available only on www.annals.org; Annals of Internal Medicine holds copyright for that article.

---

6. · 2021 · 51 cit/yr

Minimum sample size for external validation of a clinical prediction model with a binary outcome (link)

R. Riley et al.

Statistics in Medicine · May 24, 2021 · 251 citations

In prediction model research, external validation is needed to examine an existing model's performance using data independent to that for model development. Current external validation studies often suffer from small sample sizes and consequently imprecise predictive performance estimates. To address this, we propose how to determine the minimum sample size needed for a new external validation study of a prediction model for a binary outcome. Our calculations aim to precisely estimate calibration (Observed/Expected and calibration slope), discrimination (C-statistic), and clinical utility (net benefit). For each measure, we propose closed-form and iterative solutions for calculating the minimum sample size required. These require specifying: (i) target SEs (confidence interval widths) for each estimate of interest, (ii) the anticipated outcome event proportion in the validation population, (iii) the prediction model's anticipated (mis)calibration and variance of linear predictor values in the validation population, and (iv) potential risk thresholds for clinical decision-making. The calculations can also be used to inform whether the sample size of an existing (already collected) dataset is adequate for external validation. We illustrate our proposal for external validation of a prediction model for mechanical heart valve failure with an expected outcome event proportion of 0.018. Calculations suggest at least 9835 participants (177 events) are required to precisely estimate the calibration and discrimination measures, with this number driven by the calibration slope criterion, which we anticipate will often be the case. Also, 6443 participants (116 events) are required to precisely estimate net benefit at a risk threshold of 8%. Software code is provided.

---

7. · 2021 · 74 cit/yr

Second opinion needed: communicating uncertainty in medical machine learning (link)

Benjamin Kompa, Jasper Snoek, and A. Beam

NPJ Digital Medicine · Jan 5, 2021 · 388 citations

There is great excitement that medical artificial intelligence (AI) based on machine learning (ML) can be used to improve decision making at the patient level in a variety of healthcare settings. However, the quantification and communication of uncertainty for individual predictions is often neglected even though uncertainty estimates could lead to more principled decision-making and enable machine learning models to automatically or semi-automatically abstain on samples for which there is high uncertainty. In this article, we provide an overview of different approaches to uncertainty quantification and abstention for machine learning and highlight how these techniques could improve the safety and reliability of current ML systems being used in healthcare settings. Effective quantification and communication of uncertainty could help to engender trust with healthcare workers, while providing safeguards against known failure modes of current machine learning approaches. As machine learning becomes further integrated into healthcare environments, the ability to say “I'm not sure” or “I don't know” when uncertain is a necessary capability to enable safe clinical deployment.

---

8. · 2021 · 4.7 cit/yr

Incorporating uncertainty in learning to defer algorithms for safe computer-aided diagnosis (link)

Jessie Liu, B. Gallego, and S. Barbieri

Scientific Reports · Aug 17, 2021 · 22 citations

Deep neural networks are increasingly being used for computer-aided diagnosis, but erroneous diagnoses can be extremely costly for patients. We propose a learning to defer with uncertainty (LDU) algorithm which identifies patients for whom diagnostic uncertainty is high and defers them for evaluation by human experts. LDU was evaluated on the diagnosis of myocardial infarction (using discharge summaries), the diagnosis of any comorbidities (using structured data), and the diagnosis of pleural effusion and pneumothorax (using chest x-rays), and compared with ‘learning to defer without uncertainty information’ (LD) and ‘direct triage by uncertainty’ (DT) methods. LDU achieved the same F1 score as LD but deferred considerably fewer patients (e.g. 36% vs. 69% deferral rate for diagnosing pleural effusion with an F1 score of 0.96). Furthermore, even when many patients were assigned the wrong diagnosis with high confidence (e.g. for the diagnosis of any comorbidities) LDU achieved a 17% increase in F1 score, whereas DT was not applicable. Importantly, the weight of the defer loss in LDU can be easily adjusted to obtain the desired trade-off between diagnostic accuracy and deferral rate. In conclusion, LDU can readily augment any existing diagnostic network to reduce the risk of erroneous diagnoses in clinical practice.

---

9. · 2019 · 121 cit/yr

A simple, step-by-step guide to interpreting decision curve analysis (link)

A. Vickers, B. Van calster, and E. Steyerberg

Diagnostic and Prognostic Research · Oct 4, 2019 · 787 citations

Decision curve analysis is a method to evaluate prediction models and diagnostic tests that was introduced in a 2006 publication. Decision curves are now commonly reported in the literature, but there remains widespread misunderstanding of and confusion about what they mean. In this paper, we present a didactic, step-by-step introduction to interpreting a decision curve analysis and answer some common questions about the method. We argue that many of the difficulties with interpreting decision curves can be solved by relabeling the y-axis as "benefit" and the x-axis as "preference." A model or test can be recommended for clinical use if it has the highest level of benefit across a range of clinically reasonable preferences. Decision curves are readily interpretable if readers and authors follow a few simple guidelines.

---

10. · 2016 · 52 cit/yr

Assessing the Clinical Impact of Risk Prediction Models With Decision Curves: Guidance for Correct Interpretation and Appropriate Use. (link)

Kathleen F. Kerr, Marshall D. Brown, Kehao Zhu, and H. Janes

Journal of clinical oncology : official journal of the American Society of Clinical Oncology · Sep 27, 2016 · 495 citations

The decision curve is a graphical summary recently proposed for assessing the potential clinical impact of risk prediction biomarkers or risk models for recommending treatment or intervention. It was applied recently in an article in Journal of Clinical Oncology to measure the impact of using a genomic risk model for deciding on adjuvant radiation therapy for prostate cancer treated with radical prostatectomy. We illustrate the use of decision curves for evaluating clinical- and biomarker-based models for predicting a man's risk of prostate cancer, which could be used to guide the decision to biopsy. Decision curves are grounded in a decision-theoretical framework that accounts for both the benefits of intervention and the costs of intervention to a patient who cannot benefit. Decision curves are thus an improvement over purely mathematical measures of performance such as the area under the receiver operating characteristic curve. However, there are challenges in using and interpreting decision curves appropriately. We caution that decision curves cannot be used to identify the optimal risk threshold for recommending intervention. We discuss the use of decision curves for miscalibrated risk models. Finally, we emphasize that a decision curve shows the performance of a risk model in a population in which every patient has the same expected benefit and cost of intervention. If every patient has a personal benefit and cost, then the curves are not useful. If subpopulations have different benefits and costs, subpopulation-specific decision curves should be used. As a companion to this article, we released an R software package called DecisionCurve for making decision curves and related graphics.

---

11. · 2019 · 233 cit/yr

PROBAST: A Tool to Assess the Risk of Bias and Applicability of Prediction Model Studies (link)

R. Wolff et al.

Annals of Internal Medicine · Jan 1, 2019 · 1692 citations

Prediction relates to estimating the probability of something currently unknown. In the context of medical research, prediction typically concerns either diagnosis (probability of a certain condition being present but not yet detected) or prognosis (probability of an outcome developing in the future) (13). Prognosis applies not only to sick persons or those with an established diagnosis but also to, for example, pregnant women at risk for diabetes (4). Prediction research includes predictor finding studies, prediction model studies (development, validation, and extending or updating), and prediction model impact studies (1). Predictor finding studies (also known as risk factor or prognostic factor studies) aim to identify which predictors (such as age, disease stage, or biomarkers) independently contribute to the prediction of a diagnostic or prognostic outcome (1, 5). Prediction model studies aim to develop, validate, or update (for example, extend) a multivariable prediction model. A prediction model uses multiple predictors in combination to estimate probabilities to inform and often guide individual care (2, 6, 7). These models can predict an individual's probability of either currently having a particular outcome or disease (diagnostic prediction model) or having a particular outcome in the future (prognostic prediction model). Both types of model are widely used in various medical domains and settings (810), as evidenced by the large number of models developed in cancer (11, 12), neurology (13, 14), and cardiovascular disease (15). Prediction models are sometimes described as risk prediction models, predictive models, prediction indices or rules, or risk scores (2, 7). An example is QRISK2 for predicting cardiovascular risk (16). Prediction model impact studies evaluate the effect of using a model to guide patient care compared with not using such a model. They use a comparative design, such as a randomized trial, to study the model's effect on clinical decision making, patient outcomes, or costs of care (1). Systematic reviews have a key role in evidence-based medicine and the development of clinical guidelines (1719). They are considered to provide the most reliable form of evidence for the effects of an intervention or diagnostic test (20, 21). Systematic reviews of prediction models are a relatively new and evolving area but are increasingly undertaken to systematically identify, appraise, and summarize evidence on the performance of prediction models (1, 6, 22). Assessing the quality of included studies is a crucial step in any systematic review (20, 21). The QUIPS (Quality In Prognosis Studies) tool has been developed to assess risk of bias (ROB) in predictor finding (prognostic factor) studies (23). Researchers can use the revised Cochrane ROB Tool (ROB 2.0) (24) to investigate the methodological quality of prediction model impact studies that use a randomized comparative design, or ROBINS-I (Risk Of Bias In Nonrandomized Studies of Interventions) for those that use a nonrandomized comparative design (25). As more prediction model studies and systematic reviews of such studies are used as evidence for clinical guidance, a tool facilitating quality assessment for individual prediction model studies is urgently needed. We present PROBAST (Prediction model Risk Of Bias ASsessment Tool), a tool to assess the ROB and concerns regarding the applicability of diagnostic and prognostic prediction model studies. PROBAST can be used to assess studies of model development and model validation, including those updating a prediction model (Box 1 [26]). We refer to the accompanying explanation and elaboration document (27), for detailed explanations of how to use PROBAST and how to judge ROB and applicability. Box 1. Types of diagnostic and prognostic modeling studies or reports addressed by PROBAST. Adopted from the TRIPOD (Transparent Reporting of a multivariable prediction model for Individual Prognosis Or Diagnosis) and CHARMS (CHecklist for critical Appraisal and data extraction for systematic Reviews of prediction Modelling Studies) guidance (7, 26). PROBAST = Prediction model Risk Of Bias ASsessment Tool. Methods: Development of PROBAST Development of PROBAST was based on a 4-stage approach for developing health research reporting guidelines: define the scope, review the evidence base, use a Web-based Delphi procedure, and refine the tool through piloting (28). Guidelines explicitly aimed at the development of quality assessment tools were not available at the time (29). Development Stage 1: Scope and Definitions A steering group of 9 experts in prediction model studies and development of quality assessment tools agreed on key features of the desired scope of PROBAST. A panel of 38 experts with different backgrounds further refined the scope during the Web-based Delphi procedure. PROBAST was designed mainly to assess primary studies included in a systematic review. The group agreed that PROBAST would assess both risk of bias and concerns regarding applicability of a study evaluating a multivariable prediction model to be used for individualized diagnosis or prognosis. A domain-based structure was adopted, similar to that used in other ROB tools, such as ROB 2.0 (24), ROBINS-I (25), QUADAS-2 (Quality Assessment of Diagnostic Accuracy Studies 2) (30), and ROBIS (31). We agreed that PROBAST should cover primary studies that develop, validate, or update multivariable prediction models aiming to make individualized predictions of a diagnostic or prognostic outcome (Box 1). Studies that use multivariable modeling techniques to identify predictors (such as risk or prognostic factors) associated with an outcome but do not attempt to develop, validate, or update a model for making individualized predictions are not covered by PROBAST (5). Therefore, PROBAST is not intended for predictor finding studies or prediction model impact studies. Studies of diagnostic and prognostic models often use different terms for predictors and outcomes (Box 2). A multivariable prediction model is defined as any combination or equation of 2 or more predictors for estimating probability or risk for an individual (6, 7, 3234). Box 2. Differences between diagnostic and prognostic prediction model studies. PROBAST = Prediction model Risk Of Bias ASsessment Tool. Development Stage 2: Review of Evidence We used the following 3 approaches to build an evidence base to inform the development of PROBAST: identifying relevant methodological reviews in the area of prediction model research (November 2012 to January 2013), asking members of the steering group to identify relevant methodological studies (January 2013 to March 2013), and using the Delphi procedure to ask members of the wider group to identify additional evidence (February 2012 to July 2014). Identified literature was used to guide the scope and produce an initial list of signaling questions to consider for inclusion in PROBAST (1, 2, 57, 26, 3340). We grouped signaling questions into common themes to identify possible domains. Additional literature provided as part of the Web-based surveys informed development of the explanation and elaboration document. Development Stage 3: Web-Based Delphi Procedure We used a modified Delphi process to gain feedback and agreement on the scope, structure, and content of PROBAST. Web-based surveys were developed to gather structured feedback for each round. The 38-member Delphi group comprised methodological experts in prediction model research and development of quality assessment tools, experienced systematic reviewers, commissioners, and representatives of reimbursement agencies. We included various stakeholders to ensure that the views of end users, methodological experts, and decision makers were represented. The Delphi process consisted of 7 rounds. Round 1 asked about the scope of the tool, and participants agreed to focus on prediction model studies and follow a domain-based structure. Round 2 aimed to identify relevant domains and agree on which to include. The signaling questions for domains were refined in rounds 3 to 5. Respondents used a 1-to-5 Likert scale to rate each proposed signaling question for inclusion. They could also suggest rephrasing, provide supporting evidence (such as references to relevant studies), and suggest missing signaling questions. Round 6 refined the domains and introduced further optional guidance for using PROBAST. In the last round, participants received the agreed draft version of PROBAST and had the opportunity to provide any final feedback. Development Stage 4: Piloting and Refining the Tool We held 6 workshops on PROBAST at consecutive annual Cochrane Colloquia (Quebec, Canada, 2013; Hyderabad, India, 2014; Vienna, Austria, 2015; Seoul, South Korea, 2016; Cape Town, South Africa, 2017; and Edinburgh, United Kingdom, 2018). We also held numerous consecutive workshops with MSc and PhD students (for example, the master's program in epidemiology at Utrecht University [Utrecht, the Netherlands] and the Evidence-Based Health Care program at Oxford University [Oxford, United Kingdom]). In these workshops, we piloted the then-current version of PROBAST to gather feedback on practical issues associated with using the tool so that we could further refine and subsequently validate it. Finally, more than 50 review groups have already piloted PROBAST versions, including the final version, in their reviews. Topics included cancer, cardiology, endocrinology, pulmonology, and orthopedics. All feedback received from these initiatives was used to further inform the content and structure of PROBAST, wording of the signaling questions, and content of the guidance documents (27). Results: The PROBAST Tool What Does PROBAST Assess? PROBAST assesses both risk of bias and concerns regarding applicability of primary studies that developed or validated multivariable prediction models for diagnosis or prognosis (Boxes 1 and 2).

---

12. · 2020 · 118 cit/yr

External validation of prognostic models: what, why, how, when and where? (link)

Chava L. Ramspek, K. Jager, F. Dekker, C. Zoccali, and M. van Diepen

Clinical Kidney Journal · Nov 24, 2020 · 635 citations

Abstract Prognostic models that aim to improve the prediction of clinical events, individualized treatment and decision-making are increasingly being developed and published. However, relatively few models are externally validated and validation by independent researchers is rare. External validation is necessary to determine a prediction model's reproducibility and generalizability to new and different patients. Various methodological considerations are important when assessing or designing an external validation study. In this article, an overview is provided of these considerations, starting with what external validation is, what types of external validation can be distinguished and why such studies are a crucial step towards the clinical implementation of accurate prediction models. Statistical analyses and interpretation of external validation results are reviewed in an intuitive manner and considerations for selecting an appropriate existing prediction model and external validation population are discussed. This study enables clinicians and researchers to gain a deeper understanding of how to interpret model validation results and how to translate these results to their own patient population.

---

13. · 2023 · 56 cit/yr

There is no such thing as a validated prediction model (link)

B. Van calster, E. Steyerberg, L. Wynants, and M. van Smeden

BMC Medicine · Feb 24, 2023 · 174 citations

Background Clinical prediction models should be validated before implementation in clinical practice. But is favorable performance at internal validation or one external validation sufficient to claim that a prediction model works well in the intended clinical context? Main body We argue to the contrary because (1) patient populations vary, (2) measurement procedures vary, and (3) populations and measurements change over time. Hence, we have to expect heterogeneity in model performance between locations and settings, and across time. It follows that prediction models are never truly validated. This does not imply that validation is not important. Rather, the current focus on developing new models should shift to a focus on more extensive, well-conducted, and well-reported validation studies of promising models. Conclusion Principled validation strategies are needed to understand and quantify heterogeneity, monitor performance over time, and update prediction models when appropriate. Such strategies will help to ensure that prediction models stay up-to-date and safe to support clinical decision-making.

---

14. · 2006 · 211 cit/yr

Decision Curve Analysis: A Novel Method for Evaluating Prediction Models (link)

A. Vickers and E. Elkin

Medical Decision Making · Nov 1, 2006 · 4109 citations

---

15. · 2018 · 126 cit/yr

Reporting and Interpreting Decision Curve Analysis: A Guide for Investigators. (link)

B. Van calster et al.

European urology · Dec 1, 2018 · 928 citations

CONTEXT Urologists regularly develop clinical risk prediction models to support clinical decisions. In contrast to traditional performance measures, decision curve analysis (DCA) can assess the utility of models for decision making. DCA plots net benefit (NB) at a range of clinically reasonable risk thresholds.

OBJECTIVE To provide recommendations on interpreting and reporting DCA when evaluating prediction models.

EVIDENCE ACQUISITION We informally reviewed the urological literature to determine investigators' understanding of DCA. To illustrate, we use data from 3616 patients to develop risk models for high-grade prostate cancer (n=313, 9%) to decide who should undergo a biopsy. The baseline model includes prostate-specific antigen and digital rectal examination; the extended model adds two predictors based on transrectal ultrasound (TRUS).

EVIDENCE SYNTHESIS We explain risk thresholds, NB, default strategies (treat all, treat no one), and test tradeoff. To use DCA, first determine whether a model is superior to all other strategies across the range of reasonable risk thresholds. If so, that model appears to improve decisions irrespective of threshold. Second, consider if there are important extra costs to using the model. If so, obtain the test tradeoff to check whether the increase in NB versus the best other strategy is worth the additional cost. In our case study, addition of TRUS improved NB by 0.0114, equivalent to 1.1 more detected high-grade prostate cancers per 100 patients. Hence, adding TRUS would be worthwhile if we accept subjecting 88 patients to TRUS to find one additional high-grade prostate cancer or, alternatively, subjecting 10 patients to TRUS to avoid one unnecessary biopsy.

CONCLUSIONS The proposed guidelines can help researchers understand DCA and improve application and reporting.

PATIENT SUMMARY Decision curve analysis can identify risk models that can help us make better clinical decisions. We illustrate appropriate reporting and interpretation of decision curve analysis.

---

16. · 2022 · 30 cit/yr

Tackling prediction uncertainty in machine learning for healthcare (link)

Michelle Chua et al.

Nature Biomedical Engineering · Dec 29, 2022 · 98 citations

---

17. · 2024 · 50 cit/yr

Evaluation of clinical prediction models (part 3): calculating the sample size required for an external validation study (link)

R. Riley et al.

The BMJ · Jan 22, 2024 · 111 citations

> An external validation study evaluates the performance of a prediction model in new data, but many of these studies are too small to provide reliable answers. In the third article of their series on model evaluation, Riley and colleagues describe how to calculate the sample size required for external validation studies, and propose to avoid rules of thumb by tailoring calculations to the model and setting at hand.

---

18. · 2021 · 17 cit/yr

External validation of clinical prediction models: simulation-based sample size calculations were more reliable than rules-of-thumb (link)

Kym I. E. Snell et al.

Journal of Clinical Epidemiology · Feb 14, 2021 · 88 citations

> Highlights • After a clinical prediction model is developed, it is usually necessary to undertake an external validation study that examines the model's performance in new data from the same or different population. External validation studies should have an appropriate sample size, in order to estimate model performance measures precisely for calibration, discrimination and clinical utility.• Rules-of-thumb suggest at least 100 events and 100 nonevents. Such blanket guidance is imprecise, and not specific to the model or validation setting.• Our works shows that precision of performance estimates is affected by the model's linear predictor (LP) distribution, in addition to number of events and total sample size. Furthermore, sample sizes of 100 (or even 200) events and non-events can give imprecise estimates, especially for calibration.• Our new proposal uses a simulation-based sample size calculation, which accounts for the LP distribution and (mis)calibration in the validation sample, and calculates the sample size (and events) required conditional on these factors.• The approach requires the researcher to specify the desired precision for each performance measure of interest (calibration, discrimination, net benefit, etc), the model's anticipated LP distribution in the validation population, and whether or not the model is well calibrated. Guidance for how to specify these values is given, and R and Stata code is provided.

---

19. · 2021 · 65 cit/yr

Risk of bias in studies on prediction models developed using supervised machine learning techniques: systematic review (link)

C. A. Andaur Navarro et al.
The BMJ · Oct 20, 2021 · 290 citations

Abstract Objective To assess the methodological quality of studies on prediction models developed using machine learning techniques across all medical specialties. Design Systematic review. Data sources PubMed from 1 January 2018 to 31 December 2019. Eligibility criteria Articles reporting on the development, with or without external validation, of a multivariable prediction model (diagnostic or prognostic) developed using supervised machine learning for individualised predictions. No restrictions applied for study design, data source, or predicted patient related health outcomes. Review methods Methodological quality of the studies was determined and risk of bias evaluated using the prediction risk of bias assessment tool (PROBAST). This tool contains 21 signalling questions tailored to identify potential biases in four domains. Risk of bias was measured for each domain (participants, predictors, outcome, and analysis) and each study (overall). Results 152 studies were included: 58 (38%) included a diagnostic prediction model and 94 (62%) a prognostic prediction model. PROBAST was applied to 152 developed models and 19 external validations. Of these 171 analyses, 148 (87%, 95% confidence interval 81% to 91%) were rated at high risk of bias. The analysis domain was most frequently rated at high risk of bias. Of the 152 models, 85 (56%, 48% to 64%) were developed with an inadequate number of events per candidate predictor, 62 handled missing data inadequately (41%, 33% to 49%), and 59 assessed overfitting improperly (39%, 31% to 47%). Most models used appropriate data sources to develop (73%, 66% to 79%) and externally validate the machine learning based prediction models (74%, 51% to 88%). Information about blinding of outcome and blinding of predictors was, however, absent in 60 (40%, 32% to 47%) and 79 (52%, 44% to 60%) of the developed models, respectively. Conclusion Most studies on machine learning based prediction models show poor methodological quality and are at high risk of bias. Factors contributing to risk of bias include small study size, poor handling of missing data, and failure to deal with overfitting. Efforts to improve the design, conduct, reporting, and validation of such studies are necessary to boost the application of machine learning based prediction models in clinical practice. Systematic review registration PROSPERO CRD42019161764.

---

20. · 2019 · 3.3 cit/yr

Statistical inference for net benefit measures in biomarker validation studies (link)
T. Marsh, H. Janes, and M. Pepe
Biometrics · Nov 16, 2019 · 21 citations

Referral strategies based on risk scores and medical tests are commonly proposed. Direct assessment of their clinical utility requires implementing the strategy and is not possible in the early phases of biomarker research. Prior to late-phase studies, net benefit measures can be used to assess the potential clinical impact of a proposed strategy. Validation studies, in which the biomarker defines a prespecified referral strategy, are a gold standard approach to evaluating biomarker potential. Uncertainty, quantified by a confidence interval, is important to consider when deciding whether a biomarker warrants an impact study, does not demonstrate clinical potential, or that more data are needed. We establish distribution theory for empirical estimators of net benefit and propose empirical estimators of variance. The primary results are for the most commonly employed estimators of net benefit: from cohort and unmatched case-control samples, and for point estimates and net benefit curves. Novel estimators of net benefit under stratified two-phase and categorically matched case-control sampling are proposed and distribution theory developed. Results for common variants of net benefit and for estimation from right-censored outcomes are also presented. We motivate and demonstrate the methodology with examples from lung cancer research and highlight its application to study design.

---

21. · 1992 · 4.6 cit/yr

What price perfection? Calibration and discrimination of clinical prediction models. (link

G. Diamond

Journal of clinical epidemiology · 157 citations

---

22. · 2020 · 7.4 cit/yr

Deep Bayesian Gaussian processes for uncertainty estimation in electronic health records (link)

Yikuan Li et al.

Scientific Reports · Mar 23, 2020 · 45 citations

One major impediment to the wider use of deep learning for clinical decision making is the difficulty of assigning a level of confidence to model predictions. Currently, deep Bayesian neural networks and sparse Gaussian processes are the main two scalable uncertainty estimation methods. However, deep Bayesian neural networks suffer from lack of expressiveness, and more expressive models such as deep kernel learning, which is an extension of sparse Gaussian process, captures only the uncertainty from the higher-level latent space. Therefore, the deep learning model under it lacks interpretability and ignores uncertainty from the raw data. In this paper, we merge features of the deep Bayesian learning framework with deep kernel learning to leverage the strengths of both methods for a more comprehensive uncertainty estimation. Through a series of experiments on predicting the first incidence of heart failure, diabetes and depression applied to large-scale electronic medical records, we demonstrate that our method is better at capturing uncertainty than both Gaussian processes and deep Bayesian neural networks in terms of indicating data insufficiency and identifying misclassifications, with a comparable generalization performance. Furthermore, by assessing the accuracy and area under the receiver operating characteristic curve over the predictive probability, we show that our method is less susceptible to making overconfident predictions, especially for the minority class in imbalanced datasets. Finally, we demonstrate how uncertainty information derived by the model can inform risk factor analysis towards model interpretability.

---

23. · 2021 · 10 cit/yr

Estimation of required sample size for external validation of risk models for binary outcomes (link)

M. Pavlou et al.

Statistical Methods in Medical Research · Apr 21, 2021 · 50 citations

Risk-prediction models for health outcomes are used in practice as part of clinical decision-making, and it is essential that their performance be externally validated. An important aspect in the design of a validation study is choosing an adequate sample size. In this paper, we investigate the sample size requirements for validation studies with binary outcomes to estimate measures of predictive performance (C-statistic for discrimination and calibration slope and calibration in the large). We aim for sufficient precision in the estimated measures. In addition, we investigate the sample size to achieve sufficient power to detect a difference from a target value. Under normality assumptions on the distribution of the linear predictor, we obtain simple estimators for sample size calculations based on the measures above. Simulation studies show that the estimators perform well for common values of the C-statistic and outcome prevalence when the linear predictor is marginally Normal. Their performance deteriorates only slightly when the normality assumptions are violated. We also propose estimators which do not require normality assumptions but require specification of the marginal distribution of the linear predictor and require the use of numerical integration. These estimators were also seen to perform very well under marginal normality. Our sample size equations require a specified standard error (SE) and the anticipated C-statistic and outcome prevalence. The sample size requirement varies according to the prognostic strength of the model, outcome prevalence, choice of the performance measure and study objective. For example, to achieve an SE < 0.025 for the C-statistic, 60–170 events are required if the true C-statistic and outcome prevalence are between 0.64–0.85 and 0.05–0.3, respectively. For the calibration slope and calibration in the large, achieving SE < 0.15 would require 40–280 and 50–100 events, respectively. Our estimators may also be used for survival outcomes when the proportion of censored observations is high.

---

24. · 2019 · 147 cit/yr

PROBAST: A Tool to Assess Risk of Bias and Applicability of Prediction Model Studies: Explanation and Elaboration (link)

K. Moons et al.

Annals of Internal Medicine · Jan 1, 2019 · 1066 citations

Prediction models in health care aim to predict for an individual whether a particular outcome, such as disease, is present (diagnostic models) or whether it will occur in the future (prognostic models) (16). Diagnostic models can be used to refer patients for further testing, to initiate treatment, or to inform patients. Prognostic models can be used to aid decisions about preventive lifestyle changes, therapeutic interventions, or monitoring strategies or to stratify risk in randomized trial design and analysis (7, 8). Potential users of prediction models include health care professionals, policymakers, guideline developers, patients, and the general public. The medical literature contains thousands of studies developing and validating prediction models and often has numerous prediction models for the same target population and outcomes. For example, more than 60 models address breast cancer prognosis (9), more than 250 exist in obstetrics (10), and nearly 800 predict outcomes in patients with cardiovascular disease (11). This proliferation of prediction models will increase further with the growth of personalized or precision medicine. Systematic reviews are considered the most reliable form of evidence when addressing randomized therapeutic studies and studies of diagnostic test accuracy (12). In the era of personalized and precision medicine, interest in systematic reviews of prediction model studies is rapidly growing, as exemplified by the formation of the Cochrane Prognosis Methods Group to support systematic reviews of prognosis, including prediction model studies (13, 14). Guidance to facilitate systematic reviews of prediction models has been developed (Table 1), including for search strategies (15, 4143), formulation of the review question (16, 17), data extraction (16), and meta-analysis (17, 2225, 40, 44, 45). Table 1. Guidance on Conducting Systematic Reviews of Prediction Model Studies Assessment of risk of bias (ROB) is an essential step in any systematic review. Shortcomings in study design, conduct, and analysis can result in study estimates being at ROBthat is, at risk of results being flawed or distorted. When interpreting results from a systematic review, readers can draw stronger conclusions from a review based on primary studies at low ROB than from one based on studies at high or unclear ROB (46). Identifying the studies most relevant to the settings and populations targeted in the review (based on the applicability of primary studies to the review question) is also important. We therefore developed PROBAST (Prediction model Risk Of Bias ASsessment Tool) to address the lack of suitable tools designed specifically to assess ROB and applicability of primary prediction model studies. PROBAST consists of 4 domains containing 20 signaling questions to facilitate ROB assessment (39). The structure and rating system are similar to those in tools designed to assess ROB in randomized trials (revised Cochrane ROB Tool [ROB 2.0]), diagnostic accuracy studies (QUADAS-2 [Quality Assessment of Diagnostic Accuracy Studies 2]), and systematic reviews (ROBIS) (37, 47, 48). Although PROBAST was designed for use in systematic reviews of prediction model studies, it can also be used as a general tool for critical appraisal of (primary) prediction model studies. Here, we describe the rationale behind the domains and signaling questions, how to use them, and how to reach domain-level and overall judgments about ROB and applicability of primary studies to a review question. At the Web site (www.probast.org), 5 filled-in examples from across the medical field illustrate these processes. Because this is an area of active research, the tool, examples, and accompanying guidance will be updated when needed, and the latest version of PROBAST should always be downloaded from the Web site. Focus of PROBAST PROBAST is designed to assess primary studies that develop, validate, or update (for example, extend) multivariable prediction models for diagnosis or prognosis (Boxes 1 and 2). A multivariable prediction model is defined as any combination or equation of 2 or more predictors (such as age, sex, symptoms, signs, disease stage, or biomarkers) for estimating for an individual the probability or risk of having (diagnosis) or developing (prognosis) a particular outcome (1, 4, 68, 49, 50). Other names for prediction model include risk prediction model, predictive model, prediction index or rule, and risk score (1, 38, 4951). Box 1. Types of diagnostic and prognostic modeling studies or reports addressed by PROBAST. Adopted from the TRIPOD (Transparent Reporting of a multivariable prediction model for Individual Prognosis Or Diagnosis) and CHARMS (CHecklist for critical Appraisal and data extraction for systematic Reviews of prediction Modelling Studies) guidance (8, 16). Box 2. Differences between diagnostic and prognostic prediction model studies. PROBAST = Prediction model Risk Of Bias ASsessment Tool. Diagnostic and Prognostic Models Diagnostic prediction models estimate the probability that a certain outcome, the target condition, is currently present. Diagnostic prediction model studies typically include individuals who are suspectedbut not yet knownto have the target condition. Prognostic prediction models estimate the probability that a future outcome or event will occur, such as death, disease recurrence, disease complication, or therapy response. The time period of prediction can vary from hours (for example, preoperatively predicting postoperative nausea and vomiting) to years (for example, predicting lifelong risk for a coronary event). Although many prognostic models enroll patients with an established diagnosis, this does not have to be the starting point, as seen in models for predicting development of diabetes in pregnant women (52) or of osteoporotic fractures in the general population (53). Consistent with the TRIPOD (Transparent

Reporting of a multivariable prediction model for Individual Prognosis Or Diagnosis) statement (7, 8), PROBAST thus broadly defines prognostic models as those predicting a future outcome in persons at risk for that outcome. Diagnostic and prognostic model studies often use different terms for predictors and outcomes (Box 2). The cancer literature frequently distinguishes between prognostic and predictive models, such that predictive models identify individuals with differential treatment effects (54). These types of (predictive) models are outside the scope of this article. Types of Predictors, Outcomes, and Modeling Techniques PROBAST can be used to assess any type of diagnostic or prognostic prediction model aimed at individualized predictions regardless of the predictors used; outcomes being predicted; or methods used to develop, validate, or update (for example, extend) the model. Predictors range from demographic characteristics, medical history, and physical examination results; to imaging results, electrophysiology, blood, urine, or tissue measurements, and disease stages or characteristics; to results from omics and other new biological measurements. Predictors are also called covariates, risk indicators, prognostic factors, determinants, index test results, or independent variables (4, 68, 49, 50, 55, 56, 57). PROBAST distinguishes between candidate predictors and predictors included in the final model (57). Candidate predictors are variables considered potentially predictive of the outcome presence (diagnosis) or occurrence (prognosis)that is, all those evaluated in the study regardless of whether they are included in the final multivariable model. PROBAST primarily addresses prediction models for binary and time-to-event outcomes because these are the most common in medicine. However, the tool can also be used to assess models predicting nonbinary outcomes, such as continuous scores (for example, pain scores or cholesterol levels) or categorical outcomes (for example, the Glasgow Coma Scale). Almost all PROBAST signaling questions apply equally to continuous and categorical outcomes, except questions addressing number of outcome events per predictor and certain measures of model performance (such as the c-statistic), which are not relevant to continuous outcomes. Prediction models usually involve regression modeling techniques, such as logistic regression or survival models. Prediction models may also be developed or validated using nonregression techniques, such as neural networks, random forests, or support vector machines. As the use of routine care (and big) data increases, additional modeling techniques are becoming more common, including machine and artificial learning models. The main differences between studies using regression and other types of prediction modeling include the methods of data analysis; nonregression development models can often have greater risks of overfitting when data are sparse, and the potential lack of transparency can affect the applicability and usability of the models. In the section on tailoring PROBAST with additional signaling questions, we provide guidance about how PROBAST can be adapted to address other types of outcomes and modeling techniques. Types of Review Questions PROBAST can be used to assess different types of systematic review questions. For some review questions, all prediction model studies are relevant (including both development and validation), but for other questions only validation studies are relevant. Box 3 gives examples of potential review questions for both prognostic and diagnostic prediction models where PROBAST is applicable. CHARMS (CHecklist for critical Appraisal and data extraction for systematic Reviews of prediction Modelling Studies) and Table 2 provide explicit guidance on how to frame a focused question for reviews of prediction model studies (16, 17). Box 3. Examples of systematic review questions for which PROBAST is suitable. There are various different questions that systematic reviews of

25. · 2022 · 9.1 cit/yr

Calibrated Selective Classification (link)

Adam Fisch, T. Jaakkola, and R. Barzilay

Trans. Mach. Learn. Res. · Aug 25, 2022 · 33 citations

Selective classification allows models to abstain from making predictions (e.g., say"I don't know") when in doubt in order to obtain better effective accuracy. While typical selective models can be effective at producing more accurate predictions on average, they may still allow for wrong predictions that have high confidence, or skip correct predictions that have low confidence. Providing calibrated uncertainty estimates alongside predictions – probabilities that correspond to true frequencies – can be as important as having predictions that are simply accurate on average. However, uncertainty estimates can be unreliable for certain inputs. In this paper, we develop a new approach to selective classification in which we propose a method for rejecting examples with"uncertain"uncertainties. By doing so, we aim to make predictions with {well-calibrated} uncertainty estimates over the distribution of accepted examples, a property we call selective calibration. We present a framework for learning selectively calibrated models, where a separate selector network is trained to improve the selective calibration error of a given base model. In particular, our work focuses on achieving robust calibration, where the model is intentionally designed to be tested on out-of-domain data. We achieve this through a training strategy inspired by distributionally robust optimization, in which we apply simulated input perturbations to the known, in-domain training data. We demonstrate the empirical effectiveness of our approach on multiple image classification and lung cancer risk assessment tasks.

---

26. · 2018 · 19 cit/yr

Direct Uncertainty Prediction for Medical Second Opinions (link)

M. Raghu et al.

International Conference on Machine Learning · Jul 4, 2018 · 146 citations

The issue of disagreements amongst human experts is a ubiquitous one in both machine learning and medicine. In medicine, this often corresponds to doctor disagreements on a patient diagnosis. In this work, we show that machine learning models can be trained to give uncertainty scores to data instances that might result in high expert disagreements. In particular, they can identify patient cases that would benefit most from a medical second opinion. Our central methodological finding is that Direct Uncertainty Prediction (DUP), training a model to predict an uncertainty score directly from the raw patient features, works better than Uncertainty Via Classification, the two-step process of training a classifier and postprocessing the output distribution to give an uncertainty score. We show this both with a theoretical result, and on extensive evaluations on a large scale medical imaging application.

---

27. · 2020 · 16 cit/yr

Trust Issues: Uncertainty Estimation Does Not Enable Reliable OOD Detection On Medical Tabular Data (link)

Dennis Ulmer, L. Meijerink, and G. Ciná

ML4H@NeurIPS · Nov 6, 2020 · 85 citations

When deploying machine learning models in high-stakes real-world environments such as health care, it is crucial to accurately assess the uncertainty concerning a model's prediction on abnormal inputs. However, there is a scarcity of literature analyzing this problem on medical data, especially on mixed-type tabular data such as Electronic Health Records. We close this gap by presenting a series of tests including a large variety of contemporary uncertainty estimation techniques, in order to determine whether they are able to identify out-of-distribution (OOD) patients. In contrast to previous work, we design tests on realistic and clinically relevant OOD groups, and run experiments on real-world medical data. We find that almost all techniques fail to achieve convincing results, partly disagreeing with earlier findings.

---

28. · 2023 · 0.9 cit/yr

Safe and reliable transport of prediction models to new healthcare settings without the need to collect new labeled data (link)

Rudraksh Tuwani and Andrew L. Beam

medRxiv · Dec 14, 2023 · 2 citations

How can practitioners and clinicians know if a prediction model trained at a different institution can be safely used on their patient population? There is a large body of evidence showing that small changes in the distribution of the covariates used by prediction models may cause them to fail when deployed to new settings. This specific kind of dataset shift, known as covariate shift, is a central challenge to implementing existing prediction models in new healthcare environments. One solution is to collect additional labels in the target population and then fine tune the prediction model to adapt it to the characteristics of the new healthcare setting, which is often referred to as localization. However, collecting new labels can be expensive and time-consuming. To address these issues, we recast the core problem of model transportation in terms of uncertainty quantification, which allows one to know when a model trained in one setting may be safely used in a new healthcare environment of interest. Using methods from conformal prediction, we show how to transport models safely between different settings in the presence of covariate shift, even when all one has access to are covariates from the new setting of interest (e.g. no new labels). Using this approach, the model returns a prediction set that quantifies its uncertainty and is guaranteed to contain the correct label with a user-specified probability (e.g. 90%), a property that is also known as coverage. We show that a weighted conformal inference procedure based on density ratio estimation between the source and target populations can produce prediction sets with the correct level of coverage on real-world data. This allows users to know if a model's predictions can be trusted on their population without the need to collect new labeled data.

---

29. · 2018 · 12 cit/yr

Joint Active Feature Acquisition and Classification with Variable-Size Set Encoding (link)

Hajin Shim, Sung Ju Hwang, and Eunho Yang

Neural Information Processing Systems · Dec 5, 2018 · 88 citations

We consider the problem of active feature acquisition where the goal is to sequentially select the subset of features in order to achieve the maximum prediction performance in the most cost-effective way at test time. In this work, we formulate this active feature acquisition as a jointly learning problem of training both the classifier (environment) and the RL agent that decides either to `stop and predict'` `or`collect a new feature' at test time, in a cost-sensitive manner. We also introduce a novel encoding scheme to represent acquired subsets of features by proposing an order-invariant set encoding at the feature level, which also significantly reduces the search space for our agent. We evaluate our model on a carefully designed synthetic dataset for the active feature acquisition as well as several medical datasets. Our framework shows meaningful feature acquisition process for diagnosis that complies with human knowledge, and outperforms all baselines in terms of prediction performance as well as feature acquisition cost.

---

30. · 2018 · 4.2 cit/yr

An Optimal Policy for Patient Laboratory Tests in Intensive Care Units (link)

Li-Fang Cheng, Niranjani Prasad, and B. Engelhardt

Pacific Symposium on Biocomputing. Pacific Symposium on Biocomputing · Aug 14, 2018 · 32 citations

Laboratory testing is an integral tool in the management of patient care in hospitals, particularly in intensive care units (ICUs). There exists an inherent trade-off in the selection and timing of lab tests between considerations of the expected utility in clinical decision-making of a given test at a specific time, and the associated cost or risk it poses to the patient. In this work, we introduce a framework that learns policies for ordering lab tests which optimizes for this trade-off. Our approach uses batch off-policy reinforcement learning with a composite reward function based on clinical imperatives, applied to data that include examples of clinicians ordering labs for patients. To this end, we develop and extend principles of Pareto optimality to improve the selection of actions based on multiple reward function components while respecting typical procedural considerations and prioritization of clinical goals in the ICU. Our experiments show that we can estimate a policy that reduces the frequency of lab tests and optimizes timing to minimize information redundancy. We also find that the estimated policies typically suggest ordering lab tests well ahead of critical onsets—such as mechanical ventilation or dialysis—that depend on the lab results. We evaluate our approach by quantifying how these policies may initiate earlier onset of treatment.

---

31. · 2022 · 0.5 cit/yr

Confidence-based laboratory test reduction recommendation algorithm (link)

Ting Huang, L. Li, E. Bernstam, and X. Jiang

BMC Medical Informatics and Decision Making · Jun 21, 2022 · 2 citations

Background We propose a new deep learning model to identify unnecessary hemoglobin (Hgb) tests for patients admitted to the hospital, which can help reduce health risks and healthcare costs. Methods We collected internal patient data from a teaching hospital in Houston and external patient data from the MIMIC III database. The study used a conservative definition of unnecessary laboratory tests, which was defined as stable (i.e., stability) and below the lower normal bound (i.e., normality). Considering that machine learning models may yield less reliable results when trained on noisy inputs containing low-quality information, we estimated prediction confidence to assess the reliability of predicted outcomes. We adopted a "select and predict" design philosophy to maximize prediction performance by selectively considering samples with high prediction confidence for recommendations. Our model accommodated irregularly sampled observational data to make full use of variable correlations (i.e., with other laboratory test values) and temporal dependencies (i.e., previous laboratory tests performed within the same encounter) in selecting candidates for training and prediction. Results The proposed model demonstrated remarkable Hgb prediction performance, achieving a normality AUC of 95.89% and a Hgb stability AUC of 95.94%, while recommending a reduction of 9.91% of Hgb tests that were deemed unnecessary. Additionally, the model could generalize well to external patients admitted to another hospital. Conclusions This study introduces a novel deep learning model with the potential to significantly reduce healthcare costs and improve patient outcomes by identifying unnecessary laboratory tests for hospitalized patients.

---

32. · 2024 · 10 cit/yr

AI-Driven Diagnostic Assistance in Medical Inquiry: Reinforcement Learning Algorithm Development and Validation (link)

Xuan Zou et al.

Journal of Medical Internet Research · Aug 23, 2024 · 17 citations

Background For medical diagnosis, clinicians typically begin with a patient's chief concerns, followed by questions about symptoms and medical history, physical examinations, and requests for necessary auxiliary examinations to gather comprehensive medical information. This complex medical investigation process has yet to be modeled by existing artificial intelligence (AI) methodologies. Objective The aim of this study was to develop an AI-driven medical inquiry assistant for clinical diagnosis that provides inquiry recommendations by simulating clinicians' medical investigating logic via reinforcement learning. Methods We compiled multicenter, deidentified outpatient electronic health records from 76 hospitals in Shenzhen, China, spanning the period from July to November 2021. These records consisted of both unstructured textual information and structured laboratory test results. We first performed feature extraction and standardization using natural language processing techniques and then used a reinforcement learning actor-critic framework to explore the rational and effective inquiry logic. To align the inquiry process with actual clinical practice, we segmented the inquiry into 4 stages: inquiring about symptoms and medical history, conducting physical examinations, requesting auxiliary examinations, and terminating the inquiry with a diagnosis. External validation was conducted to validate the inquiry logic of the AI model. Results This study focused on 2 retrospective inquiry-and-diagnosis tasks in the emergency and pediatrics departments. The emergency departments provided records of 339,020 consultations including mainly children (median age 5.2, IQR 2.6-26.1 years) with various types of upper respiratory tract infections (250,638/339,020, 73.93%). The pediatrics department provided records of 561,659 consultations, mainly of children (median age 3.8, IQR 2.0-5.7 years) with various types of upper respiratory tract infections (498,408/561,659, 88.73%). When conducting its own inquiries in both scenarios, the AI model demonstrated high diagnostic performance, with areas under the receiver operating characteristic curve of 0.955 (95% CI 0.953-0.956) and 0.943 (95% CI 0.941-0.944), respectively. When the AI model was used in a simulated collaboration with physicians, it notably reduced the average number of physicians' inquiries to 46% (6.037/13.26; 95% CI 6.009-6.064) and 43%

(6.245/14.364; 95% CI 6.225-6.269) while achieving areas under the receiver operating characteristic curve of 0.972 (95% CI 0.970-0.973) and 0.968 (95% CI 0.967-0.969) in the scenarios. External validation revealed a normalized Kendall τ distance of 0.323 (95% CI 0.301-0.346), indicating the inquiry consistency of the AI model with physicians. Conclusions This retrospective analysis of predominantly respiratory pediatric presentations in emergency and pediatrics departments demonstrated that an AI-driven diagnostic assistant had high diagnostic performance both in stand-alone use and in simulated collaboration with clinicians. Its investigation process was found to be consistent with the clinicians' medical investigation logic. These findings highlight the diagnostic assistant's promise in assisting the decision-making processes of health care professionals.

---

33. · 2024 · 1.7 cit/yr

Extended sample size calculations for evaluation of prediction models using a threshold for classification (link)

Rebecca Whittle et al.

BMC Medical Research Methodology · Jun 28, 2024 · 3 citations

When evaluating the performance of a model for individualised risk prediction, the sample size needs to be large enough to precisely estimate the performance measures of interest. Current sample size guidance is based on precisely estimating calibration, discrimination, and net benefit, which should be the first stage of calculating the minimum required sample size. However, when a clinically important threshold is used for classification, other performance measures are also often reported. We extend the previously published guidance to precisely estimate threshold-based performance measures. We have reported closed-form solutions to estimate the sample size required to target sufficiently precise estimates of accuracy, specificity, sensitivity, positive predictive value (PPV), negative predictive value (NPV), and an iterative method to estimate the sample size required to target a sufficiently precise estimate of the F1-score, in an external evaluation study of a prediction model with a binary outcome. This approach requires the user to pre-specify the target standard error and the expected value for each performance measure alongside the outcome prevalence. We describe how the sample size formulae were derived and demonstrate their use in an example. Extension to time-to-event outcomes is also considered. In our examples, the minimum sample size required was lower than that required to precisely estimate the calibration slope, and we expect this would most often be the case. Our formulae, along with corresponding Python code and updated R, Stata and Python commands (pmvalsampsize), enable researchers to calculate the minimum sample size needed to precisely estimate threshold-based performance measures in an external evaluation study. These criteria should be used alongside previously published criteria to precisely estimate the calibration, discrimination, and net-benefit.

---

34. · 2021 · 41 cit/yr

The importance of being external. methodological insights for the external validation of machine learning models in medicine (link)

F. Cabitza et al.

Computer methods and programs in biomedicine · Jul 22, 2021 · 195 citations

---

35. · 2020 · 17 cit/yr

Minimum sample size for external validation of a clinical prediction model with a continuous outcome (link)

Lucinda Archer et al.

Statistics in Medicine · Nov 4, 2020 · 92 citations

Clinical prediction models provide individualized outcome predictions to inform patient counseling and clinical decision making. External validation is the process of examining a prediction model's performance in data independent to that used for model development. Current external validation studies often suffer from small sample sizes, and subsequently imprecise estimates of a model's predictive performance. To address this, we propose how to determine the minimum sample size needed for external validation of a clinical prediction model with a continuous outcome. Four criteria are proposed, that target precise estimates of (i) R2 (the proportion of variance explained), (ii) calibration-in-the-large (agreement between predicted and observed outcome values on average), (iii) calibration slope (agreement between predicted and observed values across the range of predicted values), and (iv) the variance of observed outcome values. Closed-form sample size solutions are derived for each criterion, which require the user to specify anticipated values of the model's performance (in particular R2) and the outcome variance in the external validation dataset. A sensible starting point is to base values on those for the model development study, as obtained from the publication or study authors. The largest sample size required to meet all four criteria is the recommended minimum sample size needed in the external validation dataset. The calculations can also be applied to estimate expected precision when an existing dataset with a fixed sample size is available, to help gauge if it is adequate. We illustrate the proposed methods on a case-study predicting fat-free mass in children.

---

36. · 2022 · 22 cit/yr

Targeted validation: validating clinical prediction models in their intended population and setting (link)

M. Sperrin, R. Riley, G. Collins, and G. Martin

Diagnostic and Prognostic Research · Dec 22, 2022 · 73 citations

Clinical prediction models must be appropriately validated before they can be used. While validation studies are sometimes carefully designed to match an intended population/setting of the model, it is common for validation studies to take place with arbitrary datasets, chosen for convenience rather than relevance. We call estimating how well a model performs within the intended population/setting "targeted validation". Use of this term sharpens the focus on the intended use of a model, which may increase the applicability of developed models, avoid misleading conclusions, and reduce research waste. It also exposes that external validation may not be required when the intended population for the model matches the population used to develop the model; here, a robust internal validation may be sufficient, especially if the development dataset was large.

---

37. · 2022 · 32 cit/yr

Stability of clinical prediction models developed using statistical or machine learning methods (link)

R. Riley and G. Collins
Biometrical Journal. Biometrische Zeitschrift · Nov 2, 2022 · 111 citations

Clinical prediction models estimate an individual's risk of a particular health outcome. A developed model is a consequence of the development dataset and model-building strategy, including the sample size, number of predictors, and analysis method (e.g., regression or machine learning). We raise the concern that many models are developed using small datasets that lead to instability in the model and its predictions (estimated risks). We define four levels of model stability in estimated risks moving from the overall mean to the individual level. Through simulation and case studies of statistical and machine learning approaches, we show instability in a model's estimated risks is often considerable, and ultimately manifests itself as miscalibration of predictions in new data. Therefore, we recommend researchers always examine instability at the model development stage and propose instability plots and measures to do so. This entails repeating the model-building steps (those used to develop the original prediction model) in each of multiple (e.g., 1000) bootstrap samples, to produce multiple bootstrap models, and deriving (i) a prediction instability plot of bootstrap model versus original model predictions; (ii) the mean absolute prediction error (mean absolute difference between individuals' original and bootstrap model predictions), and (iii) calibration, classification, and decision curve instability plots of bootstrap models applied in the original sample. A case study illustrates how these instability assessments help reassure (or not) whether model predictions are likely to be reliable (or not), while informing a model's critical appraisal (risk of bias rating), fairness, and further validation requirements.

---

38. · 2021 · 2.3 cit/yr
BEDS-Bench: Behavior of EHR-models under Distributional Shift-A Benchmark (link)
A. Avati et al.
ArXiv · Jul 17, 2021 · 11 citations

Machine learning has recently demonstrated impressive progress in predictive accuracy across a wide array of tasks. Most ML approaches focus on generalization performance on unseen data that are similar to the training data (In-Distribution, or IND). However, real world applications and deployments of ML rarely enjoy the comfort of encountering examples that are always IND. In such situations, most ML models commonly display erratic behavior on Out-of-Distribution (OOD) examples, such as assigning high confidence to wrong predictions, or vice-versa. Implications of such unusual model behavior are further exacerbated in the healthcare setting, where patient health can potentially be put at risk. It is crucial to study the behavior and robustness properties of models under distributional shift, understand common failure modes, and take mitigation steps before the model is deployed. Having a benchmark that shines light upon these aspects of a model is a first and necessary step in addressing the issue. Recent work and interest in increasing model robustness in OOD settings have focused more on image modality, while the Electronic Health Record (EHR) modality is still largely under-explored. We aim to bridge this gap by releasing BEDS-Bench, a benchmark for quantifying the behavior of ML models over EHR data under OOD settings. We use two open access, de-identified EHR datasets to construct several OOD data settings to run tests on, and measure relevant metrics that characterize crucial aspects of a model's OOD behavior. We evaluate several learning algorithms under BEDS-

Bench and find that all of them show poor generalization performance under distributional shift in general. Our results highlight the need and the potential to improve robustness of EHR models under distributional shift, and BEDS-Bench provides one way to measure progress towards that goal.

---

39. · 2021 · 47 cit/yr

Decision curve analysis to evaluate the clinical benefit of prediction models. (link)

A. Vickers and F. Holland

The spine journal : official journal of the North American Spine Society · Mar 3, 2021 · 239 citations

There is increased interest in the use of prediction models to guide clinical decision-making in orthopedics. Prediction models are typically evaluated in terms of their accuracy: discrimination (area-under-the-curve [AUC] or concordance index) and calibration (a plot of predicted vs. observed risk). But it can be hard to know how high an AUC has to be in order to be "high enough" to warrant use of a prediction model, or how much miscalibration would be disqualifying. Decision curve analysis was developed as a method to determine whether use of a prediction model in the clinic to inform decision-making would do more good than harm. Here we give a brief introduction to decision curve analysis, explaining the critical concepts of net benefit and threshold probability. We briefly review some prediction models reported in the orthopedic literature, demonstrating how use of decision curves has allowed conclusions as to the clinical value of a prediction model. Conversely, papers without decision curves were unable to address questions of clinical value. We recommend increased use of decision curve analysis to evaluate prediction models in the orthopedics literature.

---

40. · 2011 · 9.6 cit/yr

Decision curve analysis revisited: overall net benefit, relationships to ROC curve analysis, and application to case-control studies (link)

V. Rousson and T. Zumbrunn

BMC Medical Informatics and Decision Making · Jun 22, 2011 · 142 citations

BackgroundDecision curve analysis has been introduced as a method to evaluate prediction models in terms of their clinical consequences if used for a binary classification of subjects into a group who should and into a group who should not be treated. The key concept for this type of evaluation is the "net benefit", a concept borrowed from utility theory.MethodsWe recall the foundations of decision curve analysis and discuss some new aspects. First, we stress the formal distinction between the net benefit for the treated and for the untreated and define the concept of the "overall net benefit". Next, we revisit the important distinction between the concept of accuracy, as typically assessed using the Youden index and a receiver operating characteristic (ROC) analysis, and the concept of utility of a prediction model, as assessed using decision curve analysis. Finally, we provide an explicit implementation of decision curve analysis to be applied in the context of case-control studies.ResultsWe show that the overall net benefit, which combines the net benefit for the treated and the untreated, is a natural alternative to the benefit achieved by a model, being invariant with respect to the coding of the outcome, and conveying a more comprehensive picture of the situation. Further, within the framework of decision curve analysis, we illustrate the important difference between the accuracy and the utility of a model, demonstrating how poor an accurate model may be in terms of its net benefit. Eventually, we expose that the application of decision curve analysis to case-control studies, where an accurate estimate of the true prevalence of a disease cannot be obtained from the data, is achieved with a few modifications to the original calculation procedure.ConclusionsWe present several interrelated extensions to decision curve analysis that will both facilitate its interpretation and broaden its potential area of application.

---

41. · 2022 · 36 cit/yr

Methodological guidance for the evaluation and updating of clinical prediction models: a systematic review (link)

M. A. Binuya, E. Engelhardt, W. Schats, M. Schmidt, and E. Steyerberg

BMC Medical Research Methodology · Dec 12, 2022 · 119 citations

Background Clinical prediction models are often not evaluated properly in specific settings or updated, for instance, with information from new markers. These key steps are needed such that models are fit for purpose and remain relevant in the long-term. We aimed to present an overview of methodological guidance for the evaluation (i.e., validation and impact assessment) and updating of clinical prediction models. Methods We systematically searched nine databases from January 2000 to January 2022 for articles in English with methodological recommendations for the post-derivation stages of interest. Qualitative analysis was used to summarize the 70 selected guidance papers. Results Key aspects for validation are the assessment of statistical performance using measures for discrimination (e.g., C-statistic) and calibration (e.g., calibration-in-the-large and calibration slope). For assessing impact or usefulness in clinical decision-making, recent papers advise using decision-analytic measures (e.g., the Net Benefit) over simplistic classification measures that ignore clinical consequences (e.g., accuracy, overall Net Reclassification Index). Commonly recommended methods for model updating are recalibration (i.e., adjustment of intercept or baseline hazard and/or slope), revision (i.e., re-estimation of individual predictor effects), and extension (i.e., addition of new markers). Additional methodological guidance is needed for newer types of updating (e.g., meta-model and dynamic updating) and machine learning-based models. Conclusion Substantial guidance was found for model evaluation and more conventional updating of regression-based models. An important development in model evaluation is the introduction of a decision-analytic framework for assessing clinical usefulness. Consensus is emerging on methods for model updating.

---

42. · 2025 · 8.6 cit/yr

Instability of the AUROC of Clinical Prediction Models (link)

Florian D van Leeuwen et al.

Statistics in Medicine · Feb 8, 2025 · 10 citations

External validations are essential to assess the performance of a clinical prediction model (CPM) before deployment. Apart from model misspecification, also differences in patient population, the standard of care, predictor definitions, and other factors influence a model's discriminative ability, as commonly quantified by the AUC (or c-statistic). We aimed to quantify the variation in AUCs across sets of external validation studies and propose ways to adjust expectations of a model's performance in a new setting.

---

43. · 2016 · 43 cit/yr
Risk Prediction with Electronic Health Records: A Deep Learning Approach (link)
Yu Cheng, Fei Wang, Ping Zhang, and Jianying Hu
SDM · Jun 30, 2016 · 417 citations

The recent years have witnessed a surge of interests in data analytics with patient Electronic Health Records (EHR). Data-driven healthcare, which aims at effective utilization of big medical data, representing the collec-tive learning in treating hundreds of millions of patients, to provide the best and most personalized care, is be-lieved to be one of the most promising directions for transforming healthcare. EHR is one of the major car-riers for make this data-driven healthcare revolution successful. There are many challenges on working directly with EHR, such as temporality, sparsity, nois-iness, bias, etc. Thus effective feature extraction, or phenotyping from patient EHRs is a key step before any further applications. In this paper, we propose a deep learning approach for phenotyping from patient EHRs. We first represent the EHRs for every patient as a temporal matrix with time on one dimension and event on the other dimension. Then we build a four-layer convolutional neural network model for extracting phenotypes and perform prediction. The first layer is composed of those EHR matrices. The second layer is a one-side convolution layer that can extract phenotypes from the first layer. The third layer is a max pooling layer introducing sparsity on the detected phenotypes, so that only those significant phenotypes will remain. The fourth layer is a fully connected softmax prediction layer. In order to incorporate the temporal smoothness of the patient EHR, we also investigated three different temporal fusion mechanisms in the model: early fusion, late fusion and slow fusion. Finally the proposed model is validated on a real world EHR data warehouse under the specific scenario of predictive modeling of chronic diseases.

---

44. · 2019 · 1.0 cit/yr
Effective Medical Test Suggestions Using Deep Reinforcement Learning (link)
Yang Chen, Kai-Fu Tang, Yu-Shao Peng, and Edward Y. Chang
ArXiv · May 30, 2019 · 7 citations

Effective medical test suggestions benefit both patients and physicians to conserve time and improve diagnosis accuracy. In this work, we show that an agent can learn to suggest effective medical tests. We formulate the problem as a stage-wise Markov decision process and propose a reinforcement learning method to train the agent. We introduce a new representation of multiple action policy along with the training method of the proposed representation. Furthermore, a new exploration scheme is proposed to accelerate the learning of disease distributions. Our experimental results demonstrate that the accuracy of disease diagnosis can be significantly improved with good medical test suggestions.

---

45. · 2023 · 7.3 cit/yr

Deep Reinforcement Learning for Cost-Effective Medical Diagnosis (link)

Zheng Yu et al.

ArXiv · Feb 20, 2023 · 23 citations

Dynamic diagnosis is desirable when medical tests are costly or time-consuming. In this work, we use reinforcement learning (RL) to find a dynamic policy that selects lab test panels sequentially based on previous observations, ensuring accurate testing at a low cost. Clinical diagnostic data are often highly imbalanced; therefore, we aim to maximize the F_1 score instead of the error rate. However, optimizing the non-concave F_1 score is not a classic RL problem, thus invalidates standard RL methods. To remedy this issue, we develop a reward shaping approach, leveraging properties of the F_1 score and duality of policy optimization, to provably find the set of all Pareto-optimal policies for budget-constrained F_1 score maximization. To handle the combinatorially complex state space, we propose a Semi-Model-based Deep Diagnosis Policy Optimization (SM-DDPO) framework that is compatible with end-to-end training and online learning. SM-DDPO is tested on diverse clinical tasks: ferritin abnormality detection, sepsis mortality prediction, and acute kidney injury diagnosis. Experiments with real-world data validate that SM-DDPO trains efficiently and identifies all Pareto-front solutions. Across all tasks, SM-DDPO is able to achieve state-of-the-art diagnosis accuracy (in some cases higher than conventional methods) with up to 85% reduction in testing cost. The code is available at [https://github.com/Zheng321/Deep-Reinforcement-Learning-for-Cost-Effective-Medical-Diagnosis].

---

46. · 2020 · 17 cit/yr

Clinical Implementation of Predictive Models Embedded within Electronic Health Record Systems: A Systematic Review (link)

Terrence C. Lee, N. Shah, Alyssa Haack, and Sally L. Baxter

Informatics (MDPI) · Jul 25, 2020 · 98 citations

Predictive analytics using electronic health record (EHR) data have rapidly advanced over the last decade. While model performance metrics have improved considerably, best practices for implementing predictive models into clinical settings for point-of-care risk stratification are still evolving. Here, we conducted a systematic review of articles describing predictive models integrated into EHR systems and implemented in clinical practice. We conducted an exhaustive database search and extracted data encompassing multiple facets of implementation. We assessed study quality and level of evidence. We obtained an initial 3393 articles for screening, from which a final set of 44 articles was included for data extraction and analysis. The most common clinical domains of implemented predictive models were related to thrombotic disorders/anticoagulation (25%) and sepsis (16%). The majority of studies were conducted in inpatient academic settings. Implementation challenges included alert fatigue, lack of training, and increased work burden on the care team. Of 32 studies that reported effects on clinical outcomes, 22 (69%) demonstrated improvement after model implementation. Overall, EHR-based predictive models offer promising results for improving clinical outcomes, although several gaps in the literature remain, and most study designs were observational. Future studies using randomized controlled trials may help improve the generalizability of findings.

---

47. · 2019 · 2.8 cit/yr

Selective prediction-set models with coverage guarantees (link)

Jean Feng, A. Sondhi, Jessica Perry, and N. Simon

ArXiv · Jun 13, 2019 · 19 citations

Though black-box predictors are state-of-the-art for many complex tasks, they often fail to properly quantify predictive uncertainty and may provide inappropriate predictions for unfamiliar data. Instead, we can learn more reliable models by letting them either output a prediction set or abstain when the uncertainty is high. We propose training these selective prediction-set models using an uncertainty-aware loss minimization framework, which unifies ideas from decision theory and robust maximum likelihood. Moreover, since black-box methods are not guaranteed to output well-calibrated prediction sets, we show how to calculate point estimates and confidence intervals for the true coverage of any selective prediction-set model, as well as a uniform mixture of K set models obtained from K-fold sample-splitting. When applied to predicting in-hospital mortality and length-of-stay for ICU patients, our model outperforms existing approaches on both in-sample and out-of-sample age groups, and our recalibration method provides accurate inference for prediction set coverage.

---

48. · 2019 · 86 cit/yr

Conformal Prediction Under Covariate Shift (link)

R. Tibshirani, R. Barber, E. Candès, and Aaditya Ramdas

Neural Information Processing Systems · Apr 1, 2019 · 604 citations

We extend conformal prediction methodology beyond the case of exchangeable data. In particular, we show that a weighted version of conformal prediction can be used to compute distribution-free prediction intervals for problems in which the test and training covariate distributions differ, but the likelihood ratio between these two distributions is known—or, in practice, can be estimated accurately with access to a large set of unlabeled data (test covariate points). Our weighted extension of conformal prediction also applies more generally, to settings in which the data satisfies a certain weighted notion of exchangeability. We discuss other potential applications of our new conformal methodology, including latent variable and missing data problems.

---

49. · 2020 · 2.5 cit/yr

CaliForest: calibrated random forest for health data (link)

Yubin Park and Joyce Ho

Proceedings of the ACM Conference on Health, Inference, and Learning · Apr 1, 2020 · 15 citations

Real-world predictive models in healthcare should be evaluated in terms of discrimination, the ability to differentiate between high and low risk events, and calibration, or the accuracy of the risk estimates. Unfortunately, calibration is often neglected and only discrimination is analyzed. Calibration is crucial for personalized medicine as they play an increasing role in the decision making process. Since random forest is a popular model for many healthcare applications, we propose CaliForest, a new calibrated random forest. Unlike existing calibration methodologies, CaliForest utilizes the out-of-bag samples to avoid the explicit construction of a calibration set. We evaluated CaliForest on two risk prediction

tasks obtained from the publicly-available MIMIC-III database. Evaluation on these binary prediction tasks demonstrates that CaliForest can achieve the same discriminative power as random forest while obtaining a better-calibrated model evaluated across six different metrics. CaliForest will be published on the standard Python software repository and the code will be openly available on Github.

---

50. · 2021 · 2.3 cit/yr

Uncertainty-Aware Time-to-Event Prediction using Deep Kernel Accelerated Failure Time Models (link)

Zhiliang Wu, Yinchong Yang, P. Fasching, and Volker Tresp

Machine Learning in Health Care · Jul 26, 2021 · 11 citations

Recurrent neural network based solutions are increasingly being used in the analysis of longitudinal Electronic Health Record data. However, most works focus on prediction accuracy and neglect prediction uncertainty. We propose Deep Kernel Accelerated Failure Time models for the time-to-event prediction task, enabling uncertainty-awareness of the prediction by a pipeline of a recurrent neural network and a sparse Gaussian Process. Furthermore, a deep metric learning based pre-training step is adapted to enhance the proposed model. Our model shows better point estimate performance than recurrent neural network based baselines in experiments on two real-world datasets. More importantly, the predictive variance from our model can be used to quantify the uncertainty estimates of the time-to-event prediction: Our model delivers better performance when it is more confident in its prediction. Compared to related methods, such as Monte Carlo Dropout, our model offers better uncertainty estimates by leveraging an analytical solution and is more computationally efficient.

Showing top 50 of 165 papers. Full details available via CSV or BibTeX export.